Typical Textures: Thermomechanical Processing (TMP) of Metals

27-750, Spring 2006

Advanced Characterization and Microstructural Analysis

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Objectives

- Introduce you to experimentally observed textures in a wide range of materials.
- Develop a taxonomy of textures based on deformation type.
- Prepare you for relating observed textures to theoretical (numerical) models of texture development, especially the *Taylor* model.
- See chapter 5 in Kocks, Tomé & Wenk.

Taxonomy

- Deformation history more significant than alloy.
- Crystal structure determines texture through slip (and twinning) characteristics.
- Alloy (and temperature) can affect textures through planarity of slip.
- Annealing (recrystallization) sometimes produces a drastic change in texture.

Why does deformation result in texture development?

- Deformation means that a body changes its shape, which is quantified by the plastic strain, ε_p.
- Plastic strain is accommodated in crystalline materials by dislocation motion, or by re-alignment of long chain molecules in polymers.

Dislocation glide → grain reorientation

- Dislocation motion at low (homologous temperatures) occurs by glide of loops on crystallographic planes in crystallographic directions: *restricted glide*.
- Restricted glide throughout the volume is equivalent to uniform shear.
- In general, shear requires lattice rotation in order to maintain grain alignment: *compatibility*

Re-orientation
→ Preferred orientation

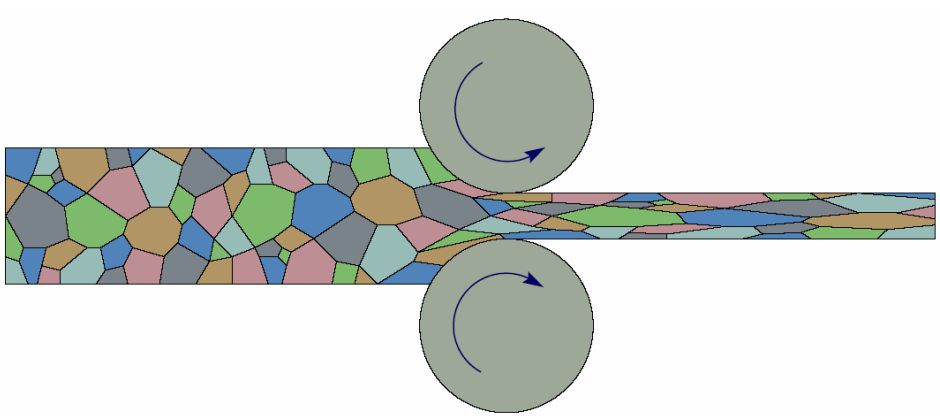
- Reorientations experienced by grains depend on the type of strain (compression versus rolling, e.g.) and the type of slip (e.g. {111}<110> in fcc).
- In general, some orientations are unstable (*f(g)* decreases) and some are stable (*f(g)* increases) with respect to the deformation imposed, hence *texture development*.

Deformation systems (typical)

Material Class	Primary System	Secondary Systems
Face-centered cubic metals	$\{111\}\langle\bar{1}10\rangle$	
Body-centered cubic metals	$\{110\}\langle\bar{1}11\rangle$	$\{123\}\langle 1\bar{1}1\rangle$ $\{112\}\langle 11\bar{1}\rangle$
Hexagonal close-packed metals (c/a>1.633) (e.g. Be, Cd, Zn and Mg)	$\{0001\}\langle 1\bar{2}10\rangle$	$\{11\bar{2}2\}\langle 1\bar{2}13\rangle$ $\{10\bar{1}1\}\langle 1\bar{2}10\rangle$
Hexagonal close-packed metals (c/a<1.633) (e.g. Zr,Ti and Hf)	$\{10\bar{1}0\}\langle 1\bar{2}10\rangle$	$\{11\bar{2}2\}\langle 1\bar{2}13\rangle$ $\{10\bar{1}1\}\langle 1\bar{2}10\rangle$
Diamond cubic (fcc) (e.g. Si, Ge and diamond)	$\{111\}\langle\bar{1}10\rangle$	
Rock Salt (fcc) (e.g. MgO, LiF, NaCl)	$\{110\}\langle\bar{1}10\rangle$	
CsCl (simple cubic)	$\{110\}\langle 001\rangle$	
Al2O3 (hexagonal)	$\{0001\}\langle 1\bar{2}10\rangle$	$\{11\bar{2}0\}\langle\bar{1}101\rangle$ $\{1\bar{1}02\}\langle\bar{1}101\rangle$
BeO (hexagonal)	$\{0001\}\langle 1\bar{2}10\rangle$	$\{10\bar{1}0\}\langle 1\bar{2}10\rangle$ $\{10\bar{1}0\}\langle 0001\rangle$

In deformed materials, texture or preferred orientation exists due to the anisotropy of slip. While slip in bcc metals generally occurs in the <111> type direction, it may be restricted to {110} planes or it may involve other planes (*T. H. Courtney, Mechanical Behavior of Materials, McGraw-Hill, New York, 1990.*)

Rolling



$$\dot{E} = \begin{pmatrix} +\Delta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\Delta \end{pmatrix}$$

Rolling ~ plane strain deformation means extension or compression in a pair of directions with zero strain in the third direction: a *multiaxial strain*.

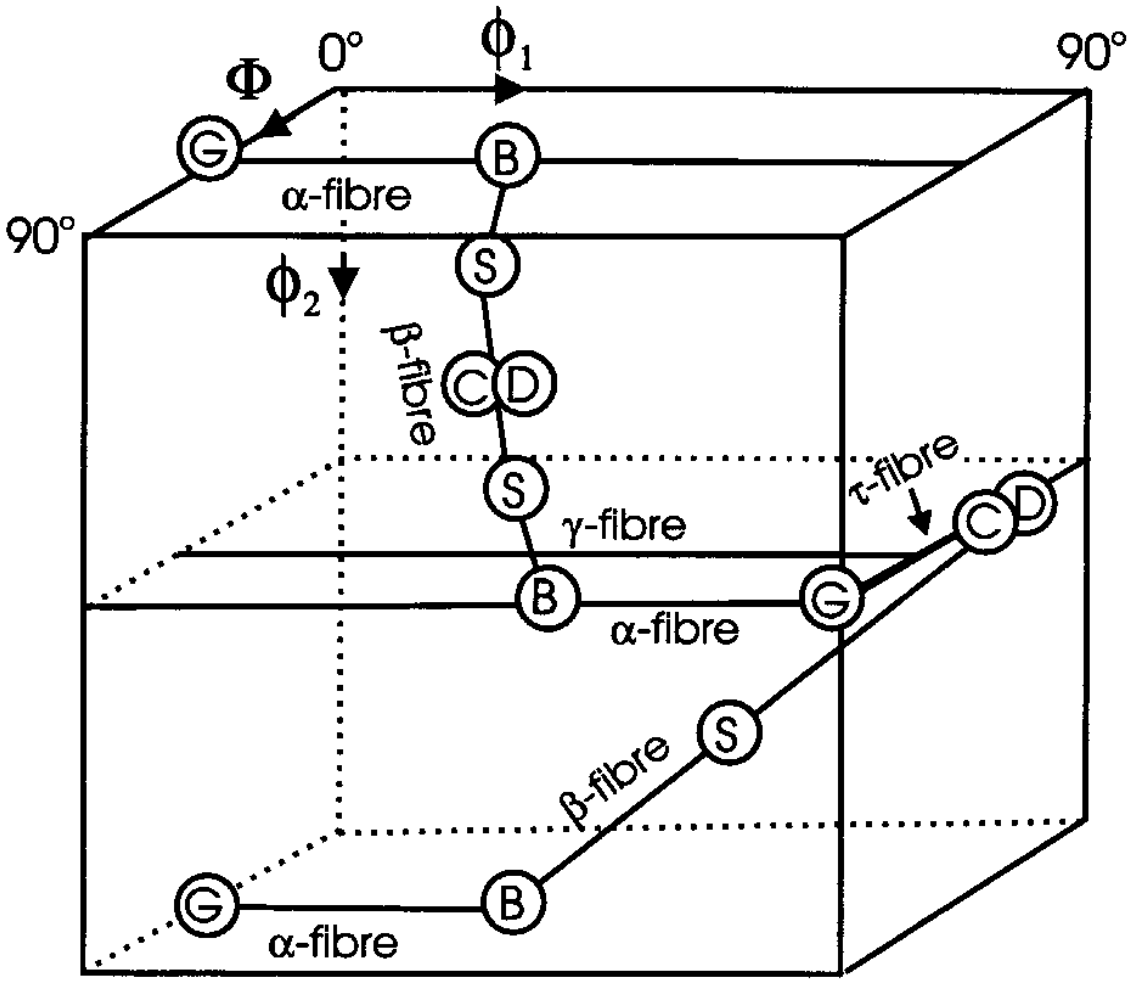
Typical rolling texture in FCC Materials

Table 4.F.2. fcc Rolling Texture Components: Euler Angles and Indices

Name	Indices	Bunge (φ_1,Φ,φ_2) RD= 1	Kocks (ψ,Θ,ϕ) RD= 1	Bunge (φ_1,Φ,φ_2) RD= 2	Kocks (ψ,Θ,ϕ) RD= 2
copper/ 1st var.	$\{112\}\langle 1\bar{1}1\rangle$	40, 65, 26	50, 65, 26	50, 65, 64	39, 66, 63
copper/ 2nd var.	$\{112\}\langle 11\bar{1}\rangle$	90, 35, 45	0, 35, 45	0, 35, 45	90, 35, 45
S3*	$\{123\}\langle 63\bar{4}\rangle$	59, 37, 27	31, 37, 27	31, 37, 63	59, 37, 63
S/ 1st var.	$(312)<0\ \bar{2}$				
S/ 2nd var.	$(312)<0\ \bar{2}$				
S/ 3rd var.	$(312)<0\ \bar{2}$				
brass/ 1st var.	$\{110\}\langle 1\bar{1}2\rangle$	35, 45, 0	55, 45, 0	55, 45, 0	35, 45, 0
brass/ 2nd var.	$\{110\}\langle 11\bar{2}\rangle$	55, 90, 45	35, 90, 45	35, 90, 45	55, 90, 45
brass/ 3rd var.	$\{110\}\langle 11\bar{2}\rangle$	35, 45, 90	55, 45, 90	55, 45, 90	35, 45, 90
Taylor	$\{4\ 4\ 11\}\langle 11\ 11\ \bar{8}\rangle$	42, 71, 20	48, 71, 20	48, 71, 70	42, 71, 70
Taylor/ 2nd var.	$\{4\ 4\ 11\}\langle 11\ 11\ 8\rangle$	90, 27, 45	0, 27, 45	0, 27, 45	90, 27, 45
Goss/ 1st var.	$\{110\}\langle 001\rangle$	0, 45, 0	90, 45, 0	90, 45, 0	0, 45, 0
Goss/ 2nd var.	$\{110\}\langle 001\rangle$	90, 90, 45	0, 90, 45	0, 90, 45	90, 90, 45
Goss/ 3rd var.	$\{110\}\langle 001\rangle$	0, 45, 90	90, 45, 90	90, 45, 90	0, 45, 90

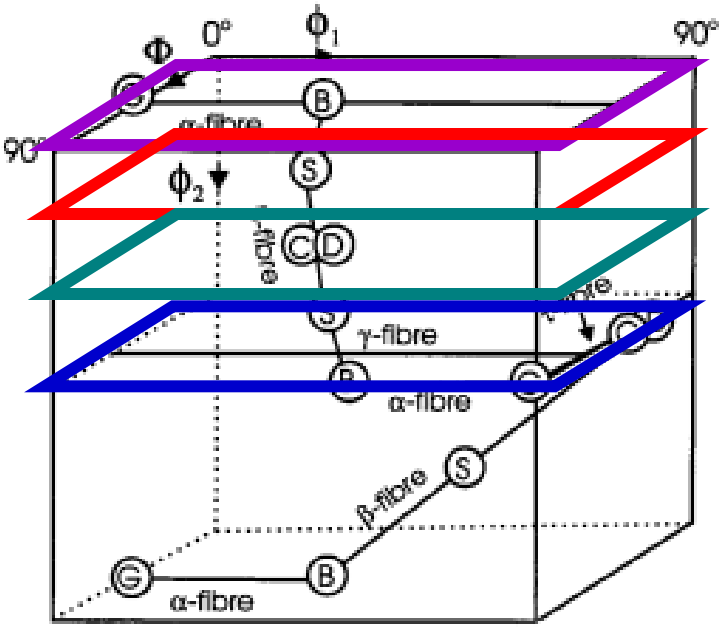
3D Representation

[Humphreys & Hatherley]

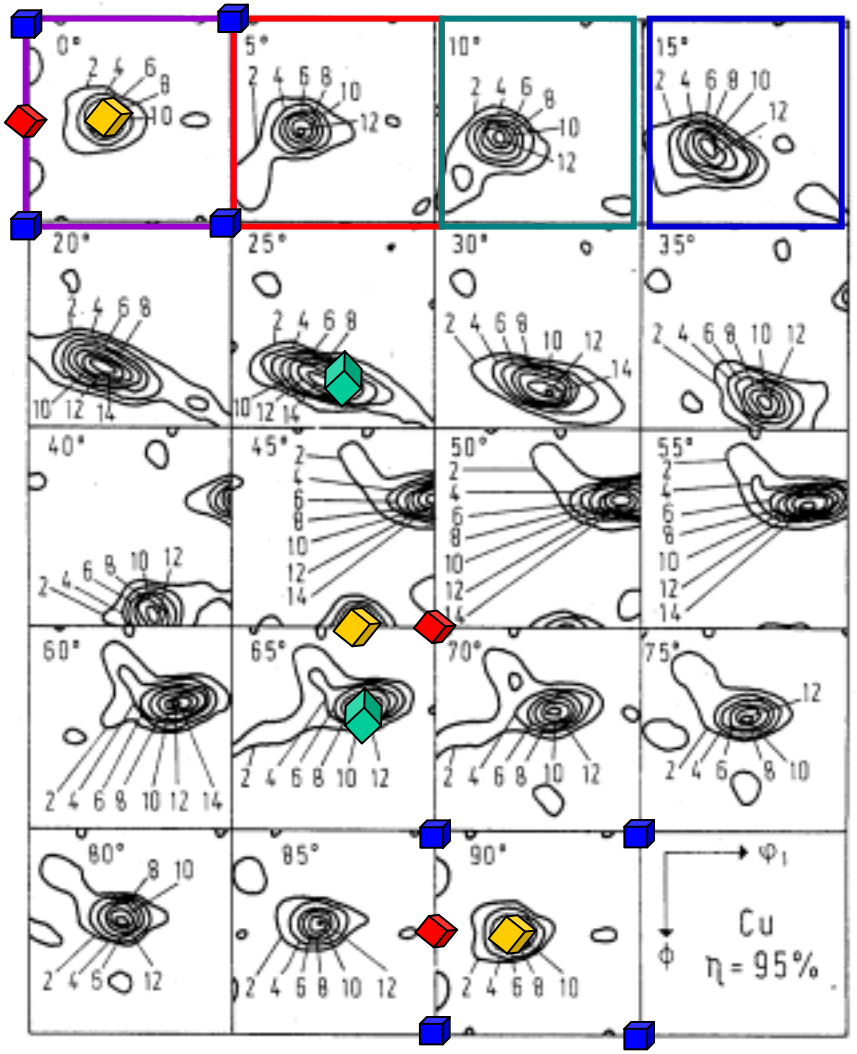


ODF Representation

Example of copper rolled to 90% reduction in thickness ($\varepsilon \sim 2.5$)



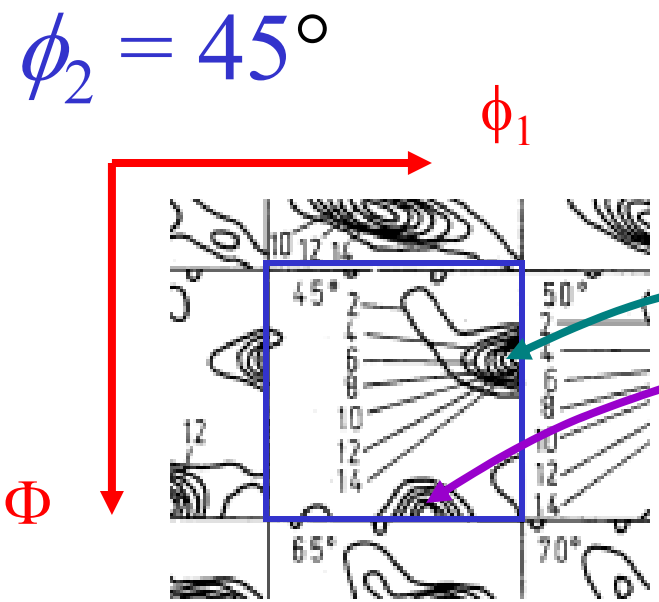
Sections are drawn as contour maps, one per value of ϕ_2 (0, 5, 10 … 90°).

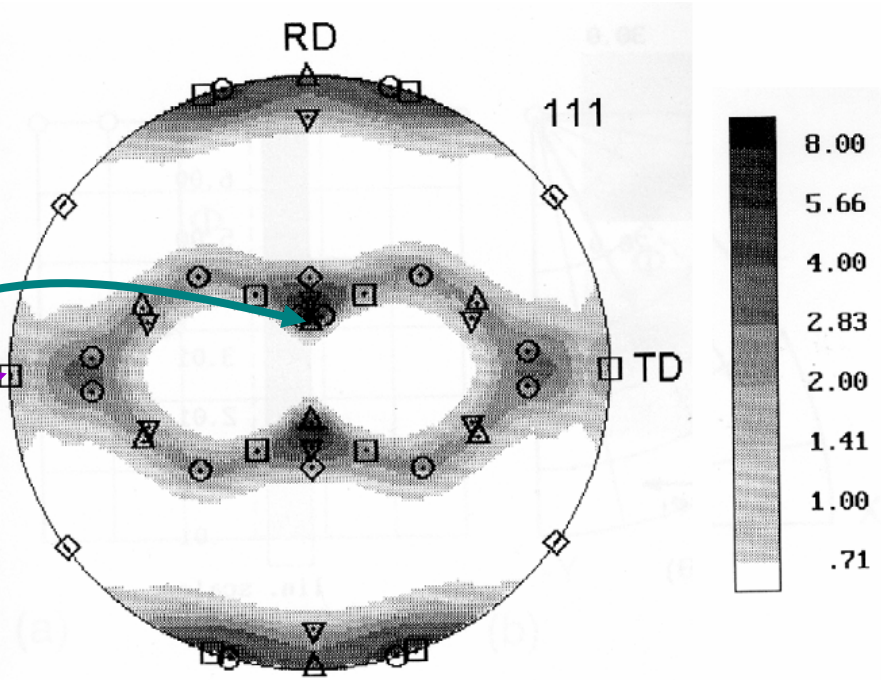


Cube Goss Brass Copper

PF Representation

$\phi_2 = 45°$





Recalculated 111 pole figure of copper rolled to 90% reduction. Δ 'Copper'; ∇ 'Dillamore'; o various 'S' components; square: 'Brass', diamond: 'Goss'.

B = Brass

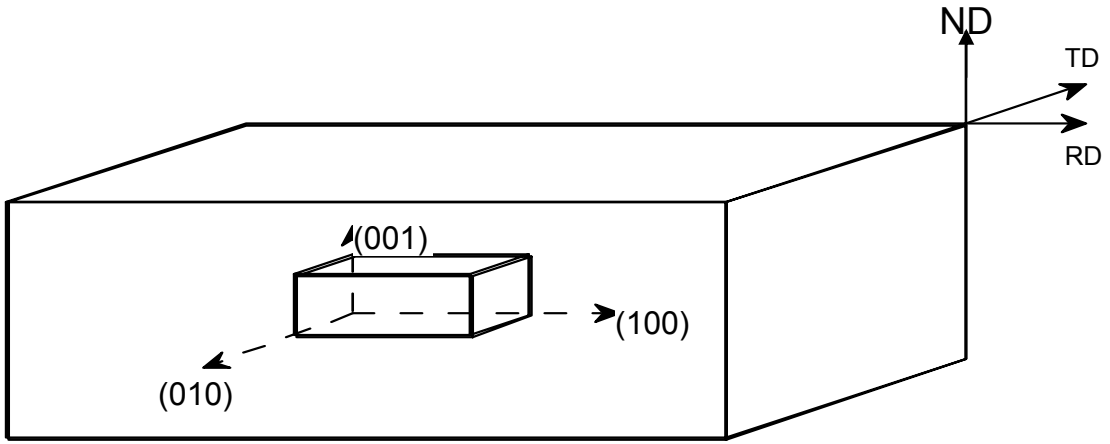
C = Copper

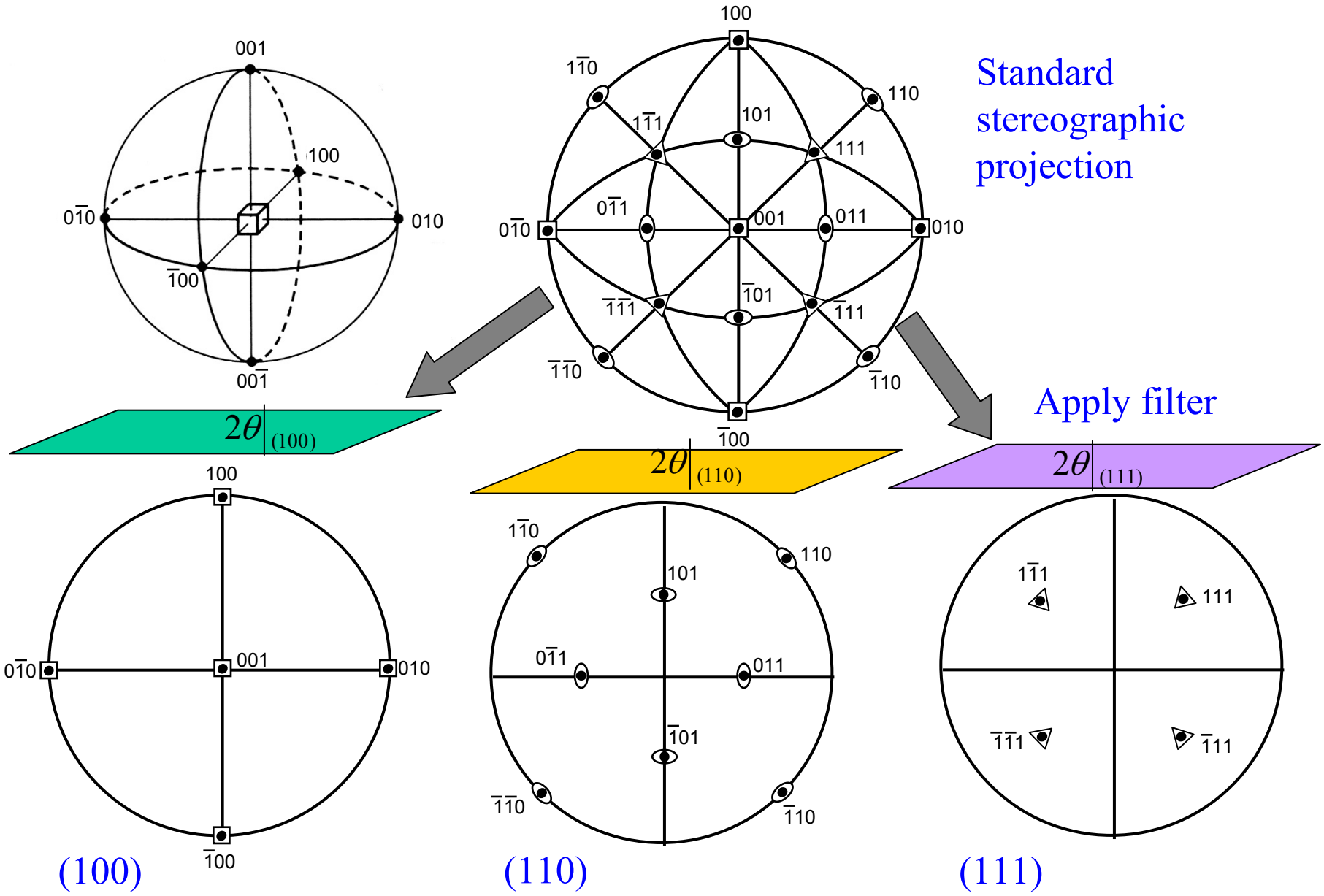
Note that any given component that is represented as a point in orientation space occurs in multiple locations in each pole figure.

PF Representation

Cube Texture {001}<100>: Cube-on-face **{ND}<RD>**

- Observed in recrystallization of *fcc* metals
- The (100) poles/orientations are parallel to the three ND, RD, and TD directions.



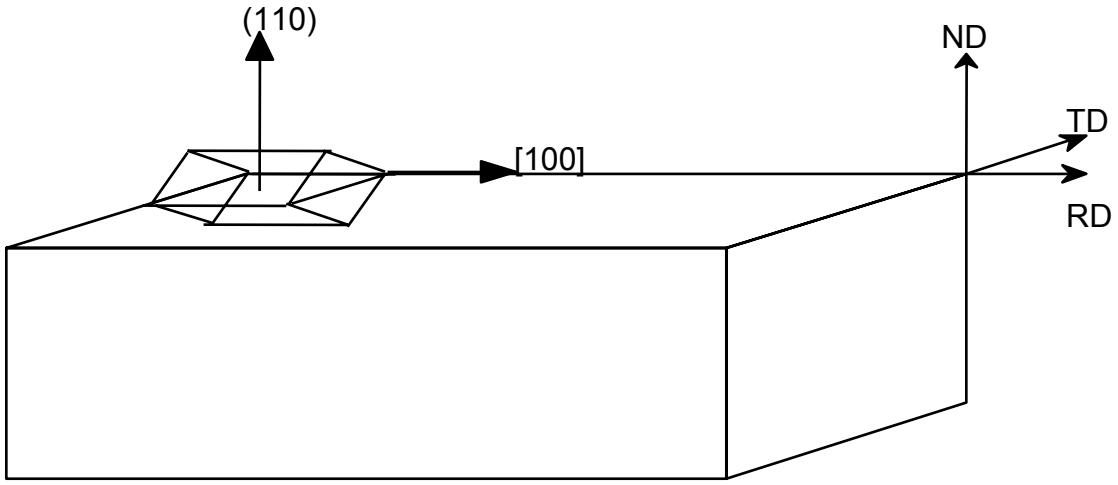


(100), (110), and (111) pole figures of CUBE {001}<100>

Goss Component: {011}<001>

Cube-on-edge

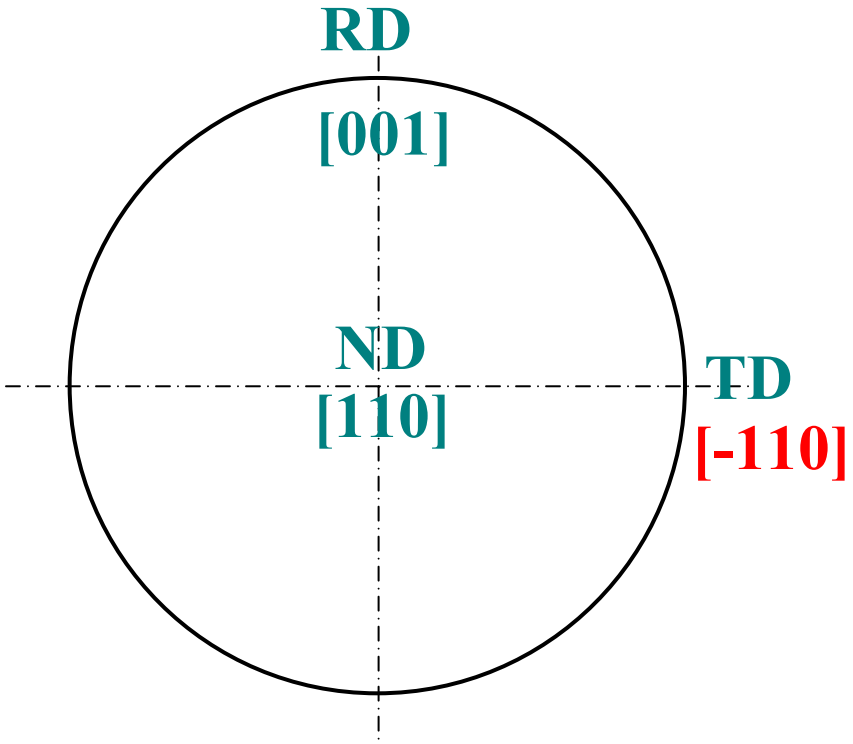
- Goss texture occurs as a recrystallization texture for FCC materials such as Brass, …
- In this case the (011) plane is oriented towards the ND and the [001] inside the (011) plane is along the RD.



PF Representation

Goss {110}<001>

{ND}<RD>



Determine TD

TD=[001] x [110]

$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$

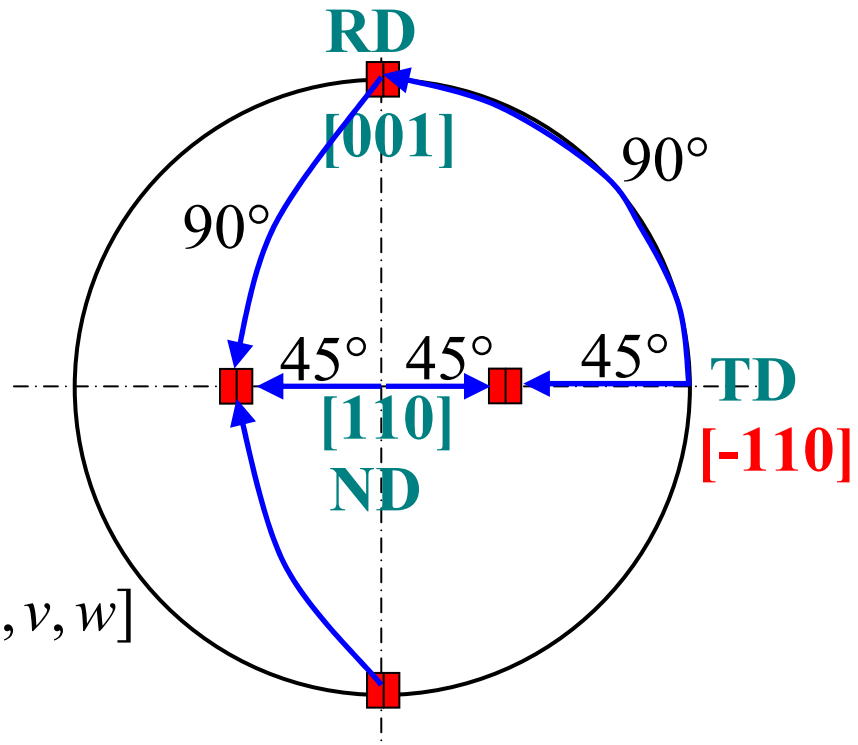
TD=[$\bar{1}$ 1 0]

Verify: $(h,k,l) \bullet (u,v,w) = 0$

Goss {110}<001>

{100} Pole Figure

- Get [100] ∢ [110]: 45°, 90°
- Get [100] ∢ [001]: 0°, 90°



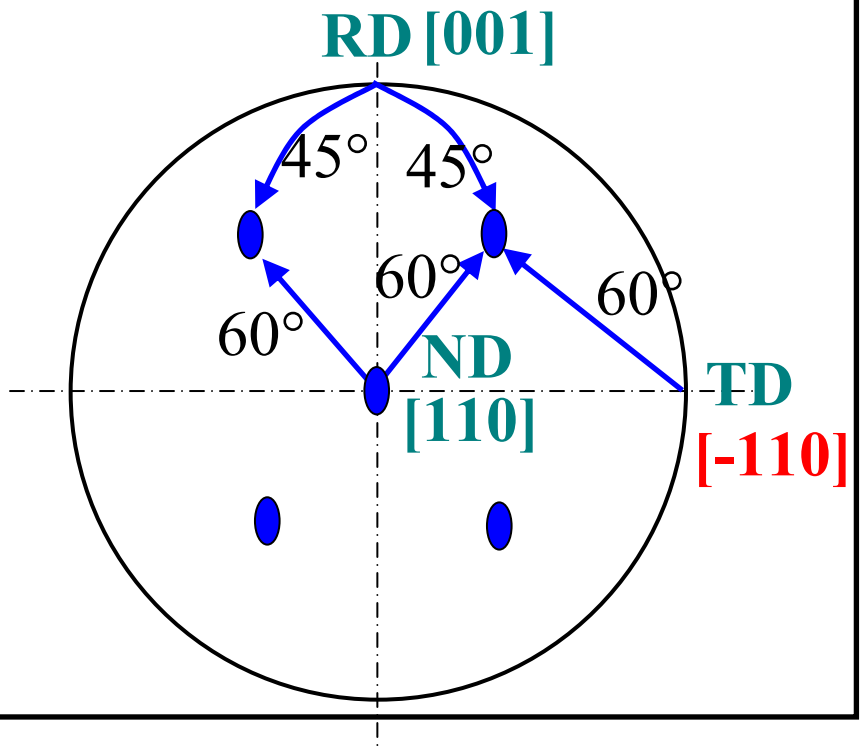
$$\cos\alpha = \frac{\mathrm{a}^{\mathrm{T}}\mathrm{b}}{\|\mathrm{a}\|_2\|\mathrm{b}\|_2}, \quad \mathrm{a} = [h,k,l]; \quad \mathrm{b} = [u,v,w]$$

$$\cos\alpha = \frac{hu + kv + lw}{\sqrt{h^2 + k^2 + l^2}\sqrt{u^2 + v^2 + w^2}}$$

Goss {110}<001>

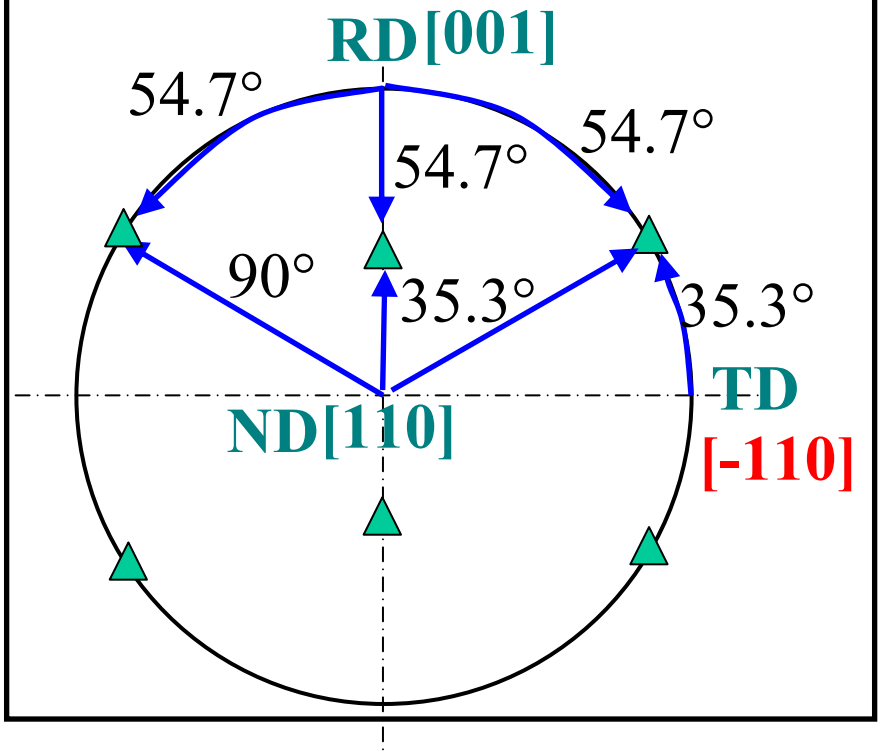
{110} Pole Figure

- Get [110] ∢ [110]: 0°, 60°, 90°
- Get [110] ∢ [001]: 45°, 90°

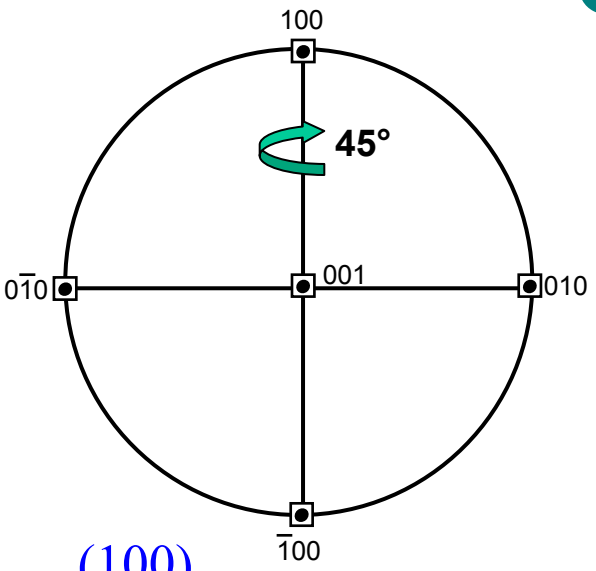


{111} Pole Figure

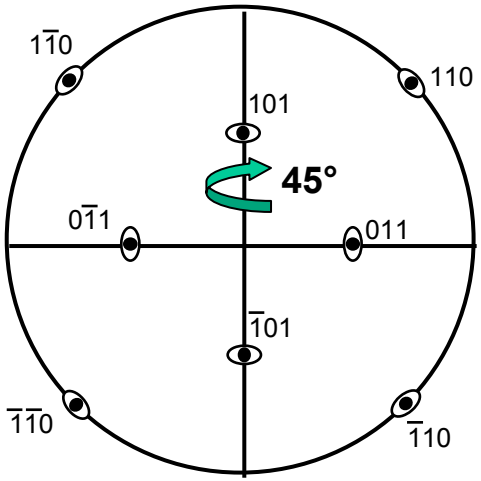
- Get [111] ∢ [110]: 35.3°, 90°
- Get [111] ∢ [001]: 54.7°



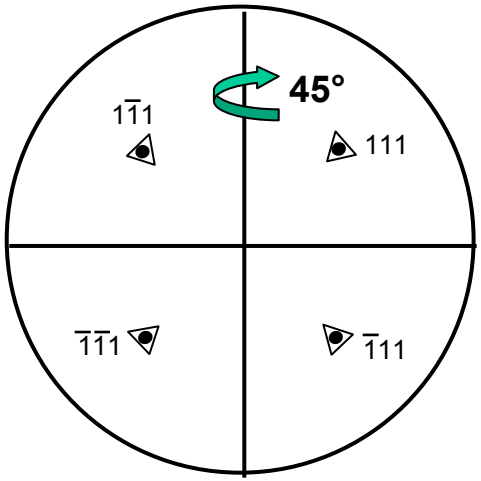
Cube {001}<100>



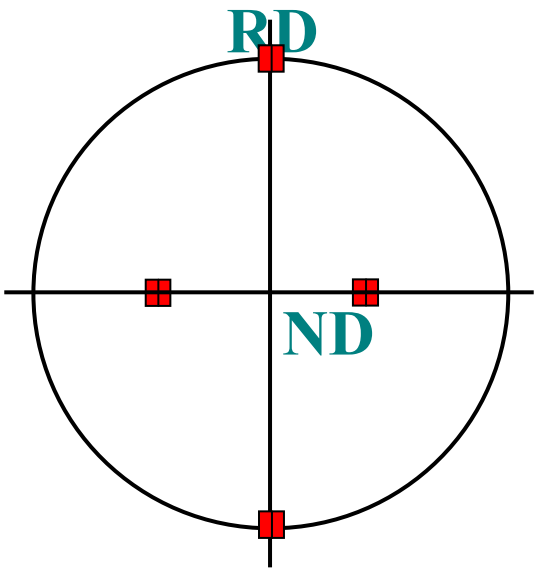
(100)

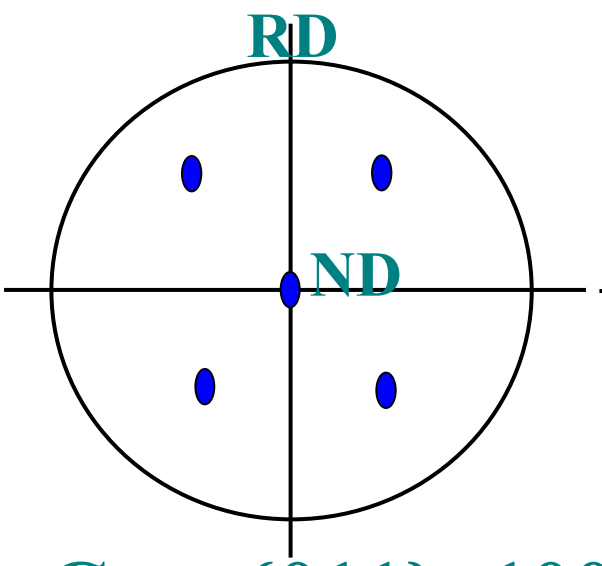


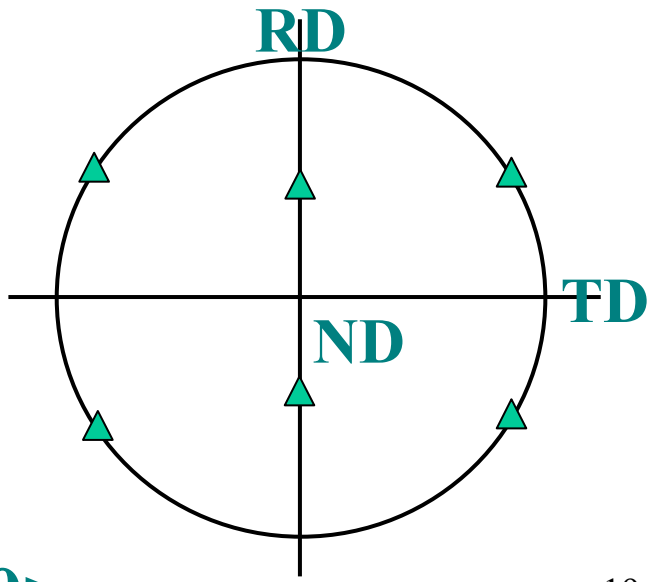
(110)



(111)







Goss {011}<100>

Recall: Axis Transformation from Axis-Angle Pair

Cube

(001)[100]

$$g = (001)[100] \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad a_{ij} = Crystal \begin{pmatrix} b_1 & t_1 & n_1 \\ b_2 & t_2 & n_2 \\ b_3 & t_3 & n_3 \end{pmatrix}^{Sample}$$

Rotation

$$a_{ij} = \delta_{ij}\cos\theta + r_i r_j (1-\cos\theta) + \sum_{k=1,3} \varepsilon_{ijk} r_k \sin\theta$$

$$a\big|_{(100),45^\circ} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \qquad ag = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

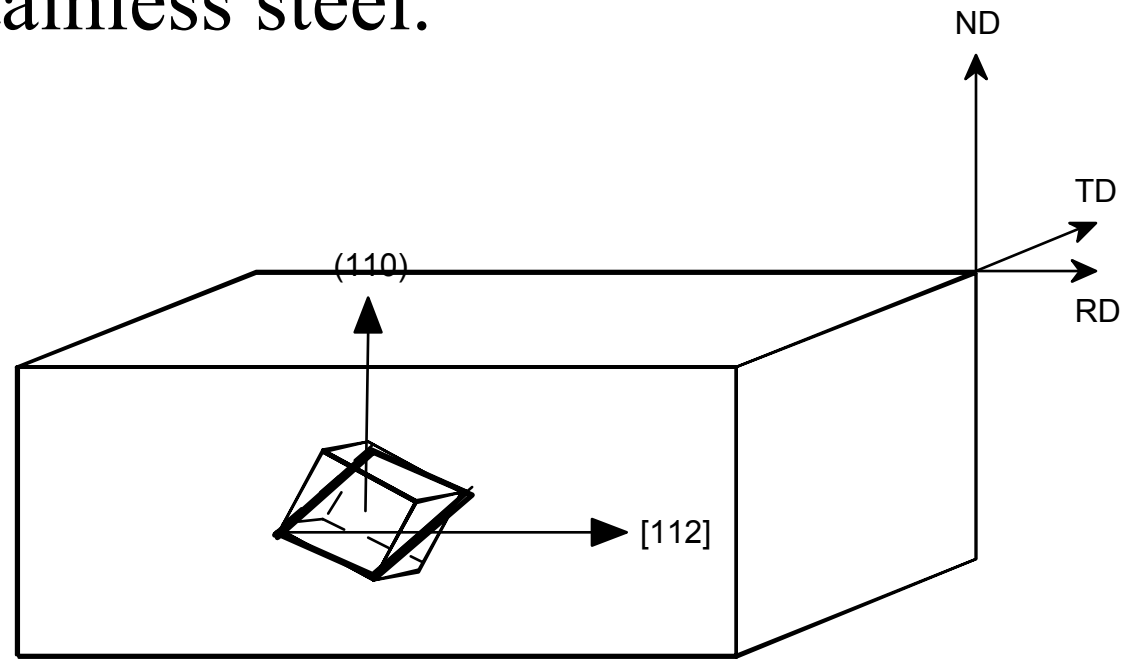
[uvw]≡RD, TD, *ND*≡*(hkl)*

GOSS

(011)[100]

Brass component {110}<112>

- Brass texture occurs as a rolling texture component for materials such as Brass, Silver, and Stainless steel.



PF Representation

Brass {110}<112>

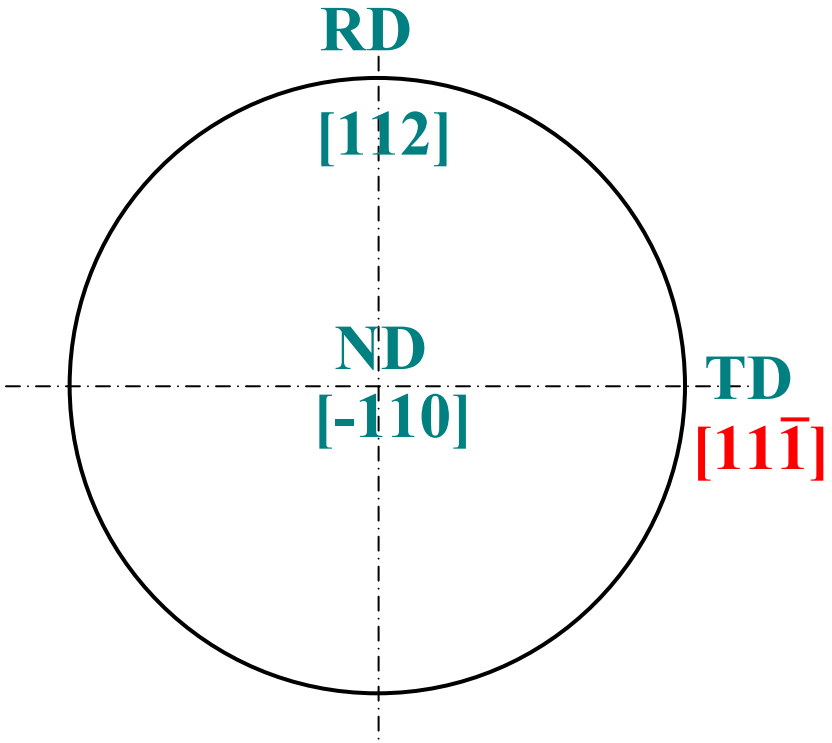
{ND}<RD>

Determine TD

TD=[-110] x [112]

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ -1 & -1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ -2 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

TD=[1 1$\bar{1}$]

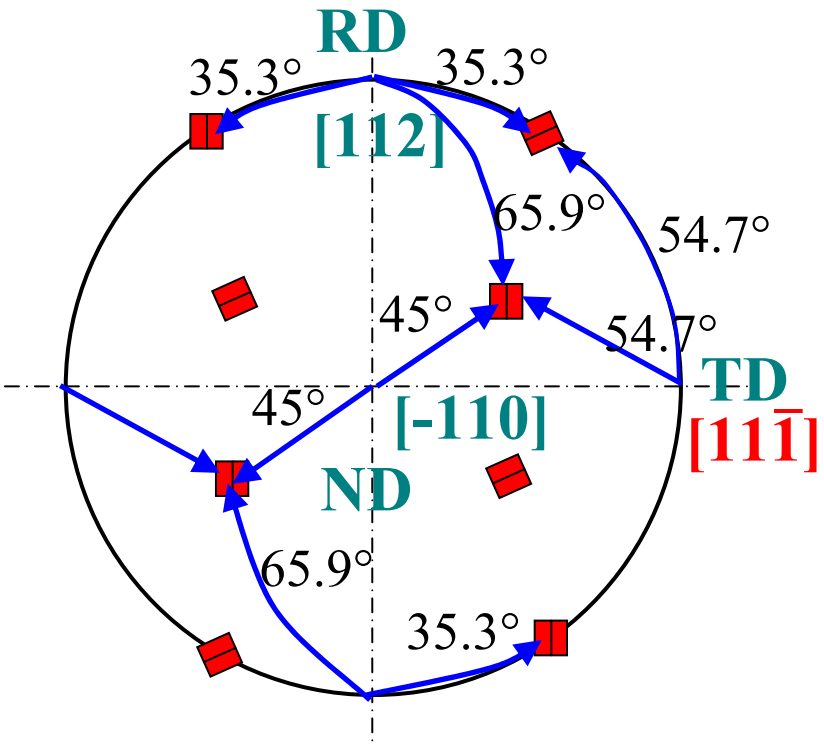


Verify: $(h,k,l)\bullet(u,v,w)=0$

Brass {110}<112>

{100} Pole Figure

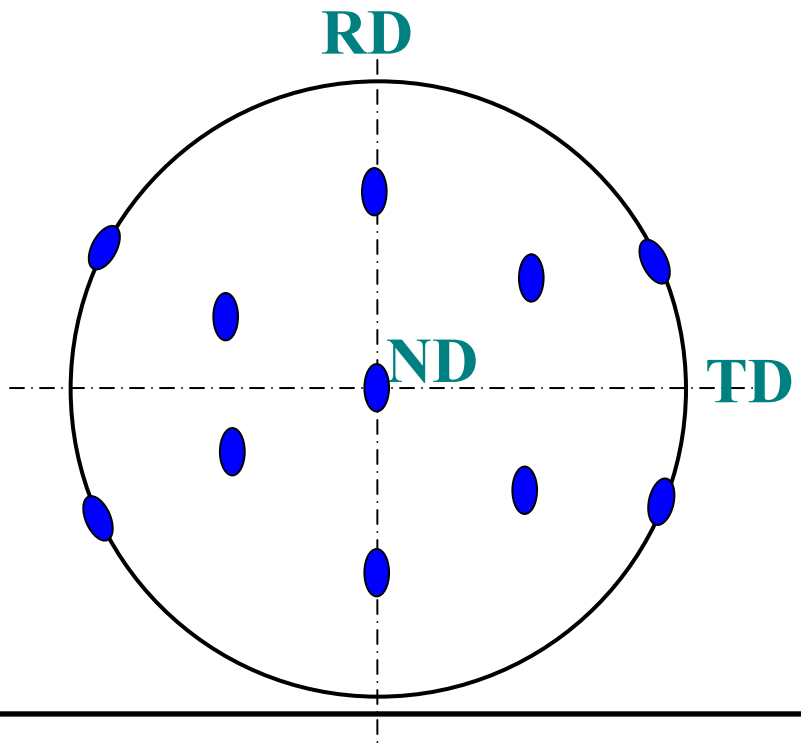
- Get [100] ∢ [110]: 45°, 90°
- Get [100] ∢ [112]: 35.3°, 65.9°
- Get [100] ∢ [111]: 54.7°



Brass {110}<112>

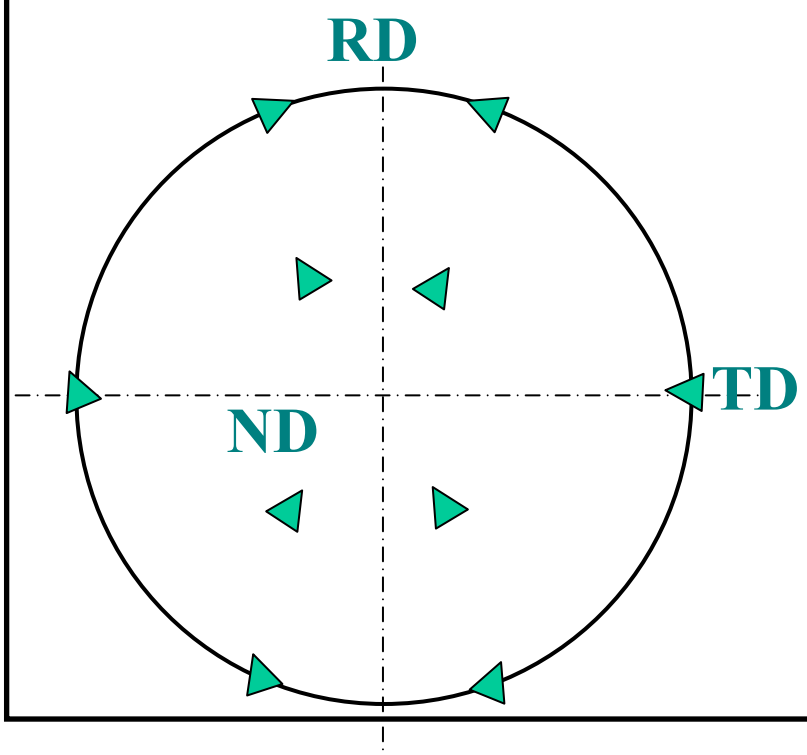
{110} Pole Figure

- Get [110] ∢ [110]: 0°, 60°, 90°
- Get [110] ∢ [112]:
  30°, 54.7°, 73.2°, 90°
- Get [110] ∢ [111]: 35.3, 90

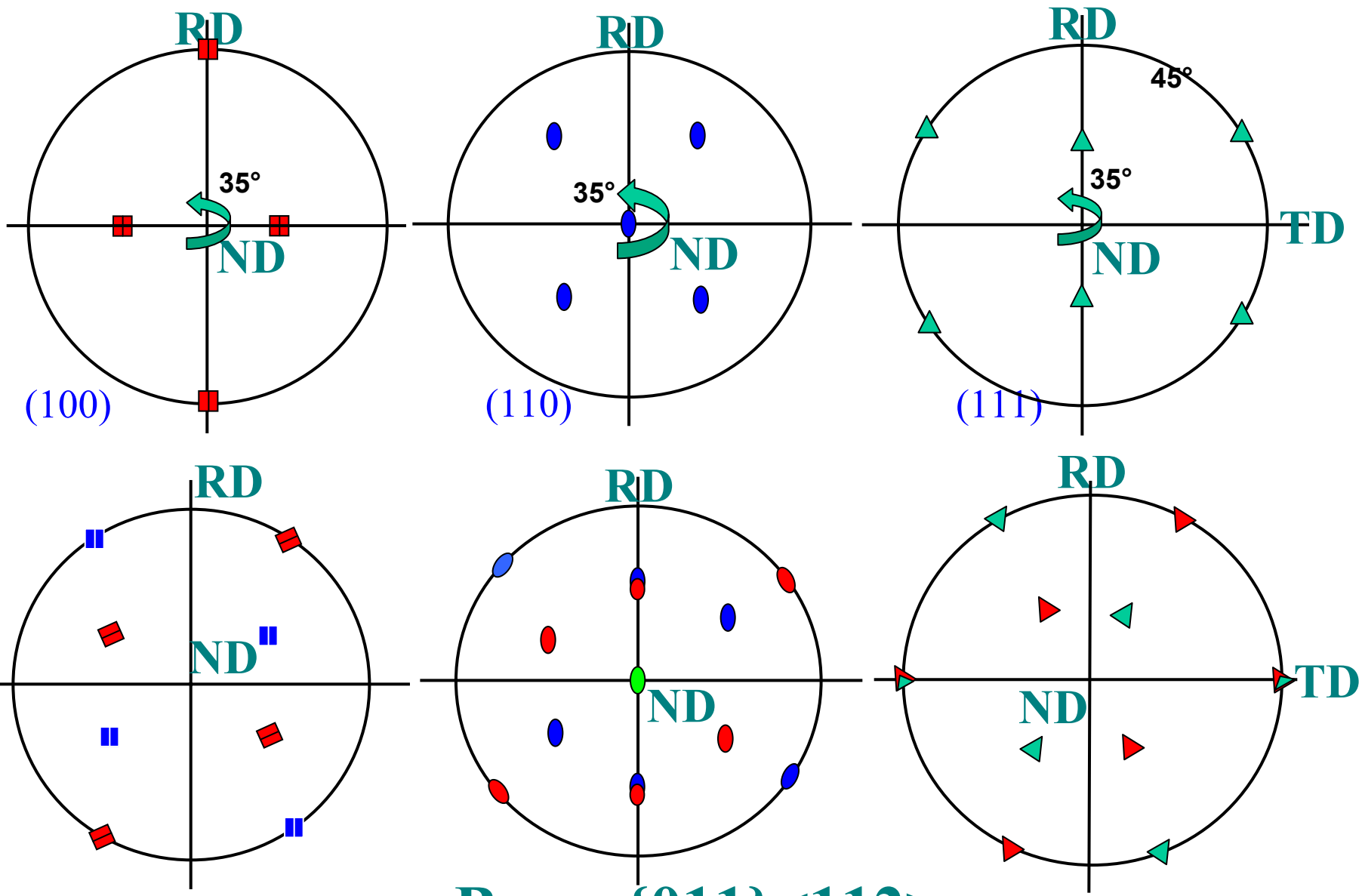


{111} Pole Figure

- Get [111] ∢ [110]: 35.3°, 90°
- Get [111] ∢ [112]:
  19.5°, 61.9°, 90°
- Get [111] ∢ [111]: 0°, 70.5°



Goss {011}<100>



Brass {011}<112>

Axis Transformation

GOSS

(011)[100]

$$g = (011)[100] \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

Rotation

$$a_{ij} = \delta_{ij}\cos\theta + r_i r_j (1-\cos\theta) + \sum_{k=1,3} \varepsilon_{ijk} r_k \sin\theta$$

$$a\big|_{(011),35^\circ} = \begin{pmatrix} \cos 35^\circ & \frac{\sin 35^\circ}{\sqrt{2}} & -\frac{\sin 35^\circ}{\sqrt{2}} \\ -\frac{\sin 35^\circ}{\sqrt{2}} & \frac{1+\cos 35^\circ}{2} & \frac{1-\cos 35^\circ}{2} \\ \frac{\sin 35^\circ}{\sqrt{2}} & \frac{1-\cos 35^\circ}{2} & \frac{1+\cos 35^\circ}{2} \end{pmatrix}$$

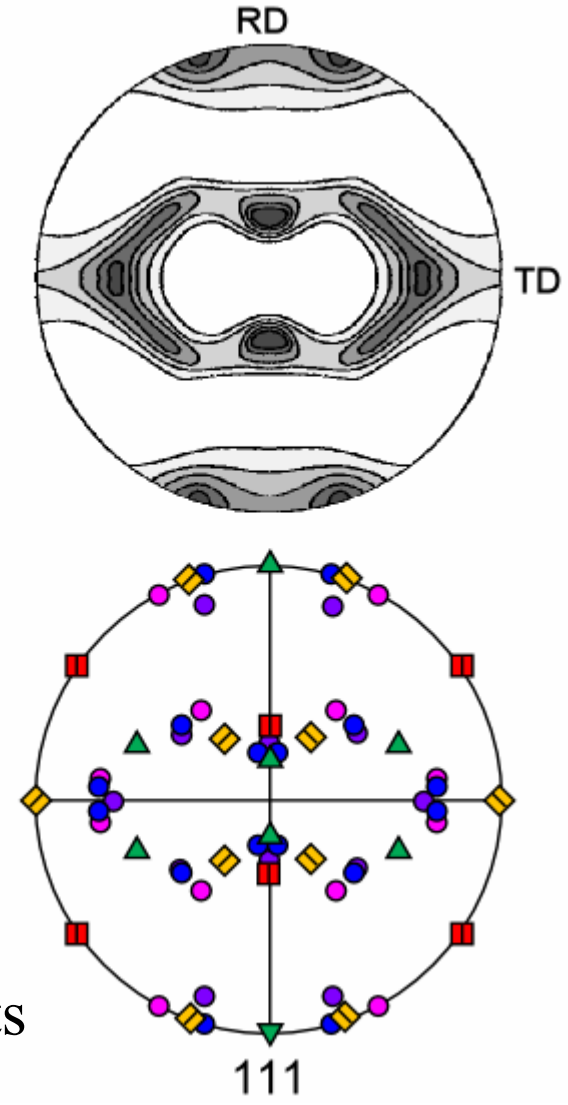
$$ag = \begin{pmatrix} \cos 35^\circ & \sin 35^\circ & 0 \\ -\frac{\sin 35^\circ}{\sqrt{2}} & \frac{\cos 35^\circ}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{\sin 35^\circ}{\sqrt{2}} & -\frac{\cos 35^\circ}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

BRASS

(011)[$2\bar{1}1$]

PF Representation

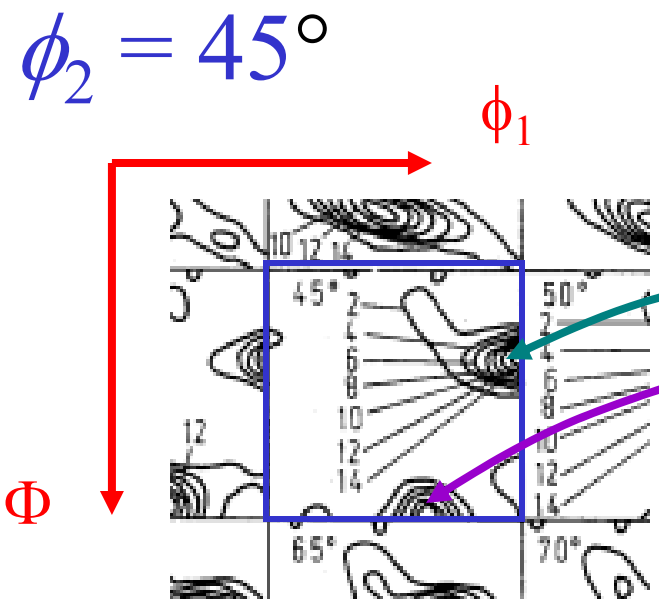
Name	Indices	Bunge ($\varphi_1, \Phi, \varphi_2$)
▲ copper	{112}<11$\bar{1}$>	90°, 35°, 45°
● S1	{124}<21$\bar{1}$>	59°, 29°, 63°
● S2	{123}<41$\bar{2}$>	47°, 37°, 63°
● S3*	{123}<63$\bar{4}$>	59°, 37°, 63°
◆ brass	{110}<$\bar{1}$12>	35°, 45°, 0°
Taylor	{4 4 11}<11 11$\bar{8}$>	7°, 71°, 70°
■ Goss	{110}<001>	0°, 45°, 0°

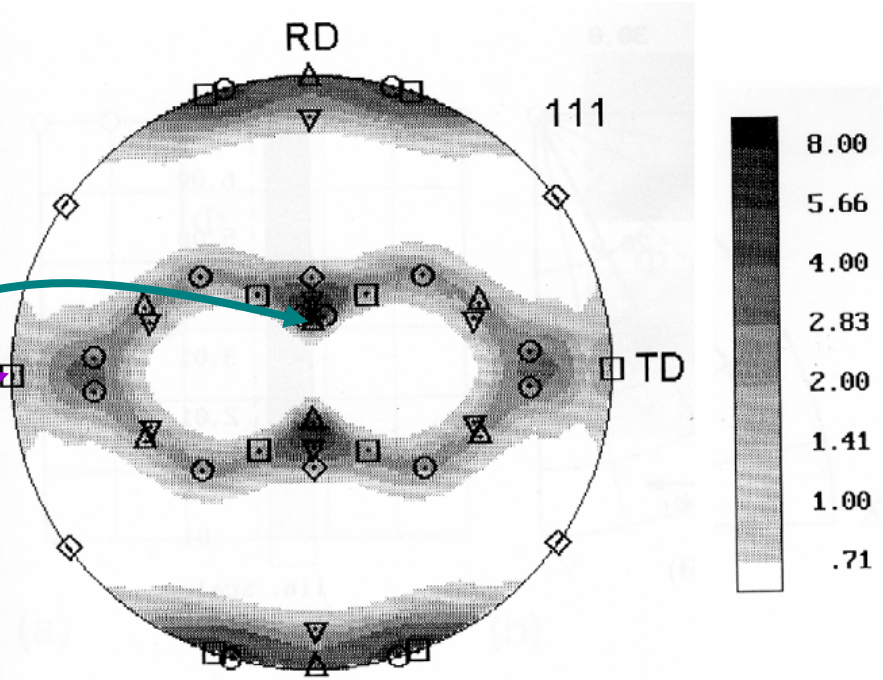


Note how very different components tend to overlap in a pole figure.

PF Representation

$\phi_2 = 45°$





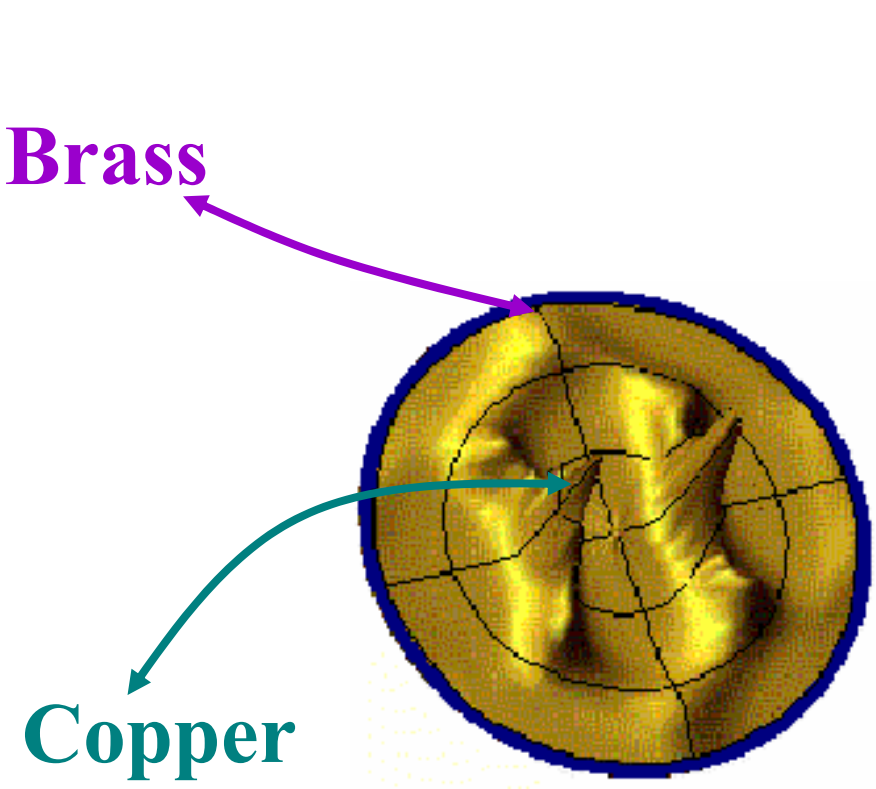
Recalculated 111 pole figure of copper rolled to 90% reduction. Δ 'Copper'; ∇ 'Dillamore'; o various 'S' components; square: 'Brass', diamond: 'Goss'.

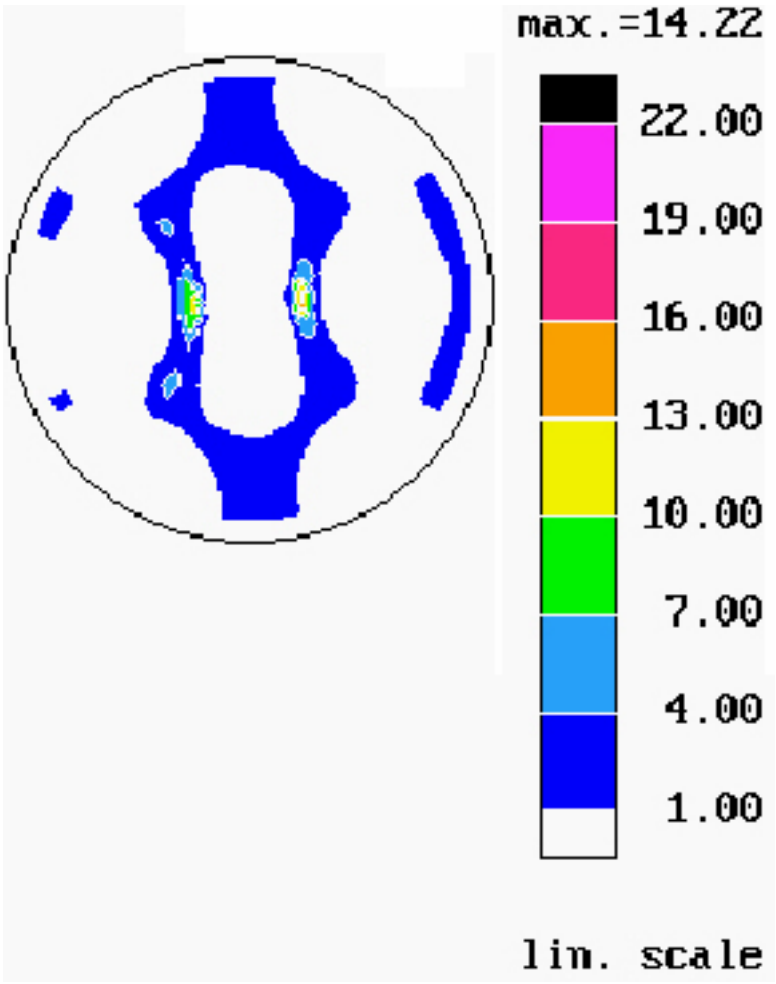
B = Brass

C = Copper

Note that any given component that is represented as a point in orientation space occurs in multiple locations in each pole figure.

PF Representation

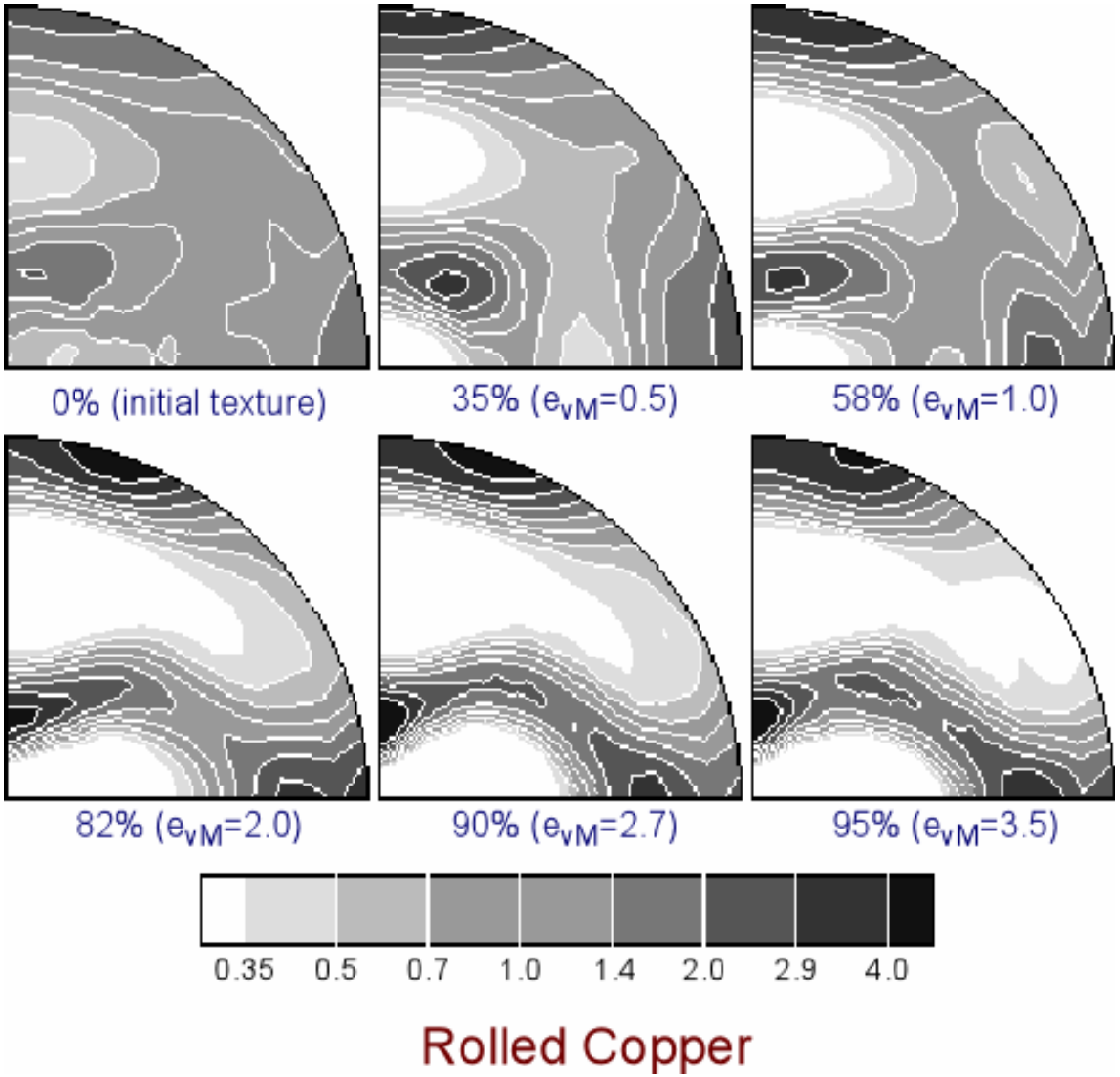




{111} pole figure of Weldalite X2095

(P. N. Kalu and D. R. Waryoba, LiMAT 2003, Honolulu.)

Effect of processing strain

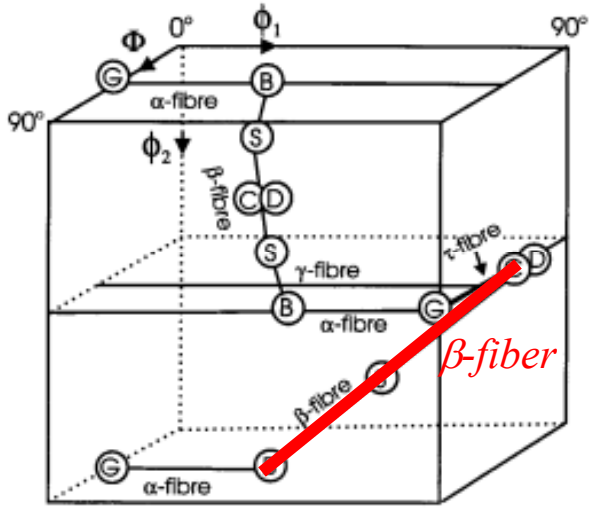


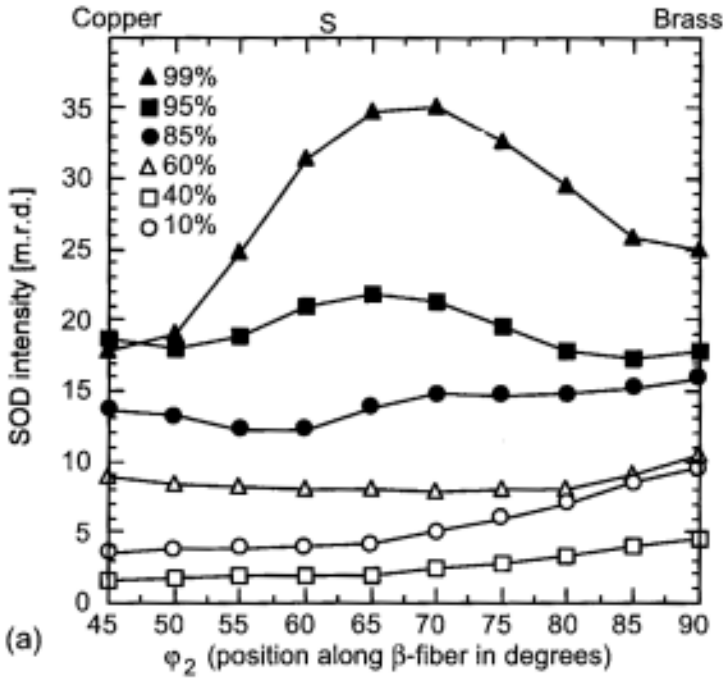
Fiber Plots:

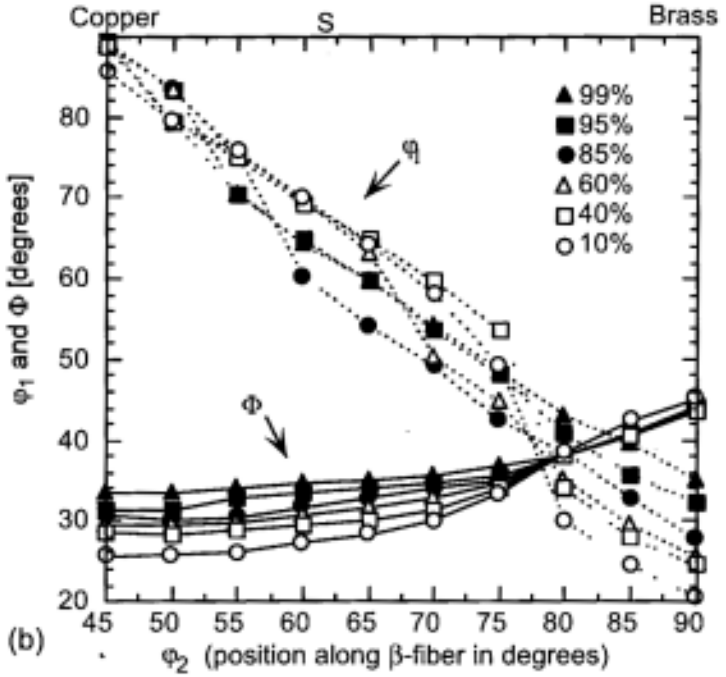
various rolling reductions:

(a) intensity versus position along the fiber

(b) angular position of intensity maximum versus position along the fiber

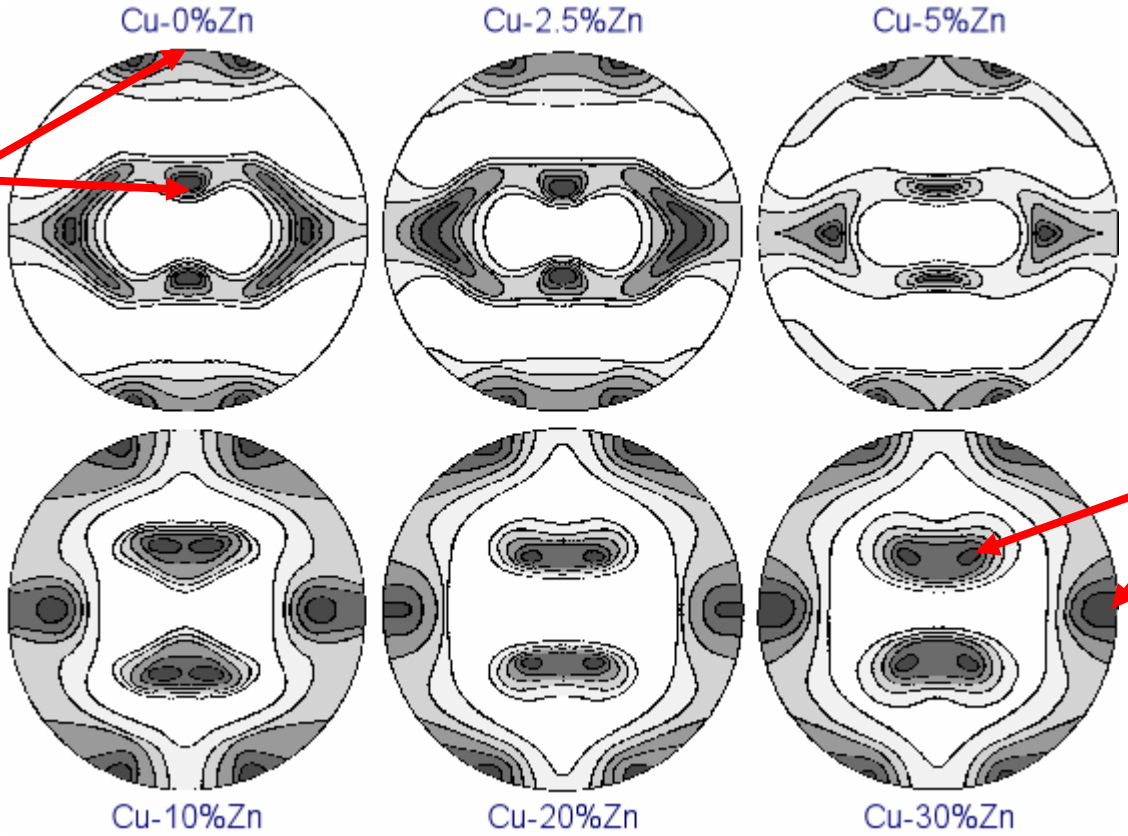






Kocks, Ch. 2

Effect of Alloying: Cu-Zn (brass); the texture transition



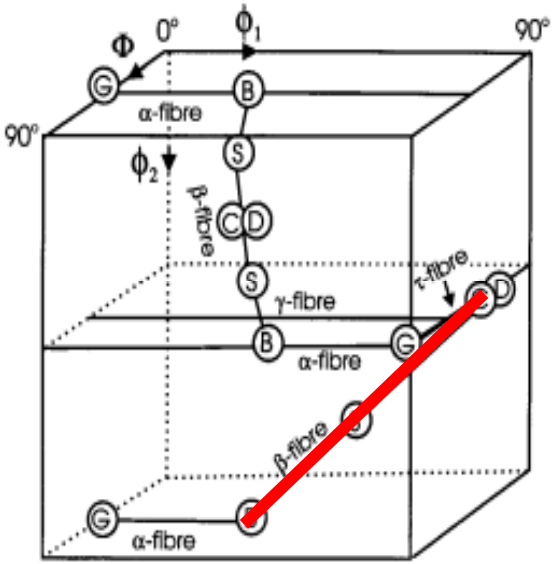
Copper component

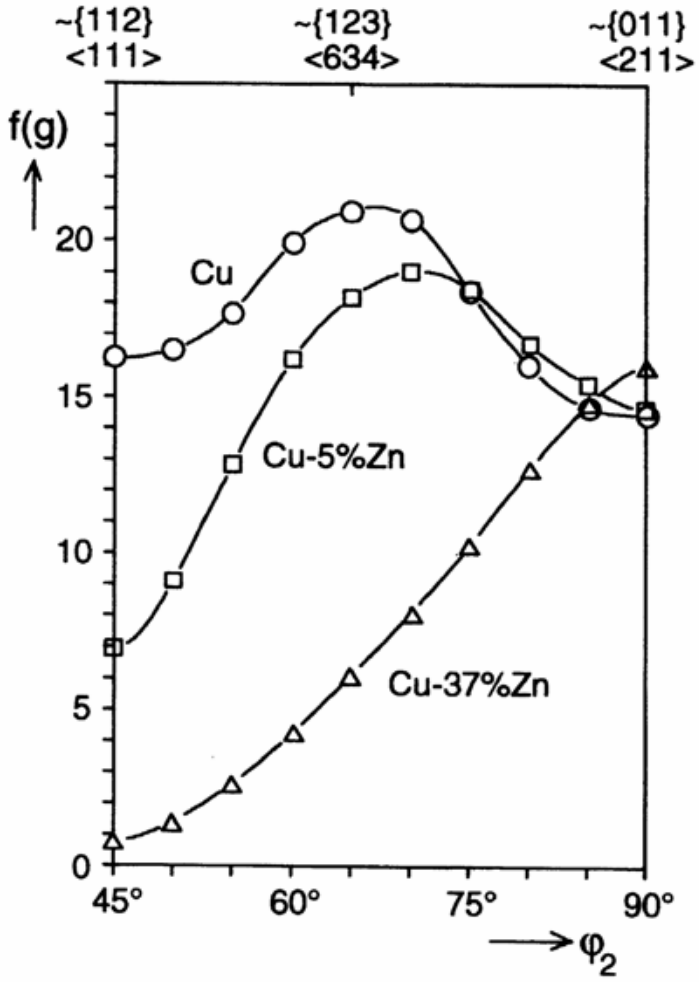
Brass

{111} Pole Figures for Copper-Zinc alloys rolled to 96% reduction

Stephens, A.S. (1968). Texture and mechanical anisotropy in the copper-zinc system. Ph.D. Thesis, University of Arizona.

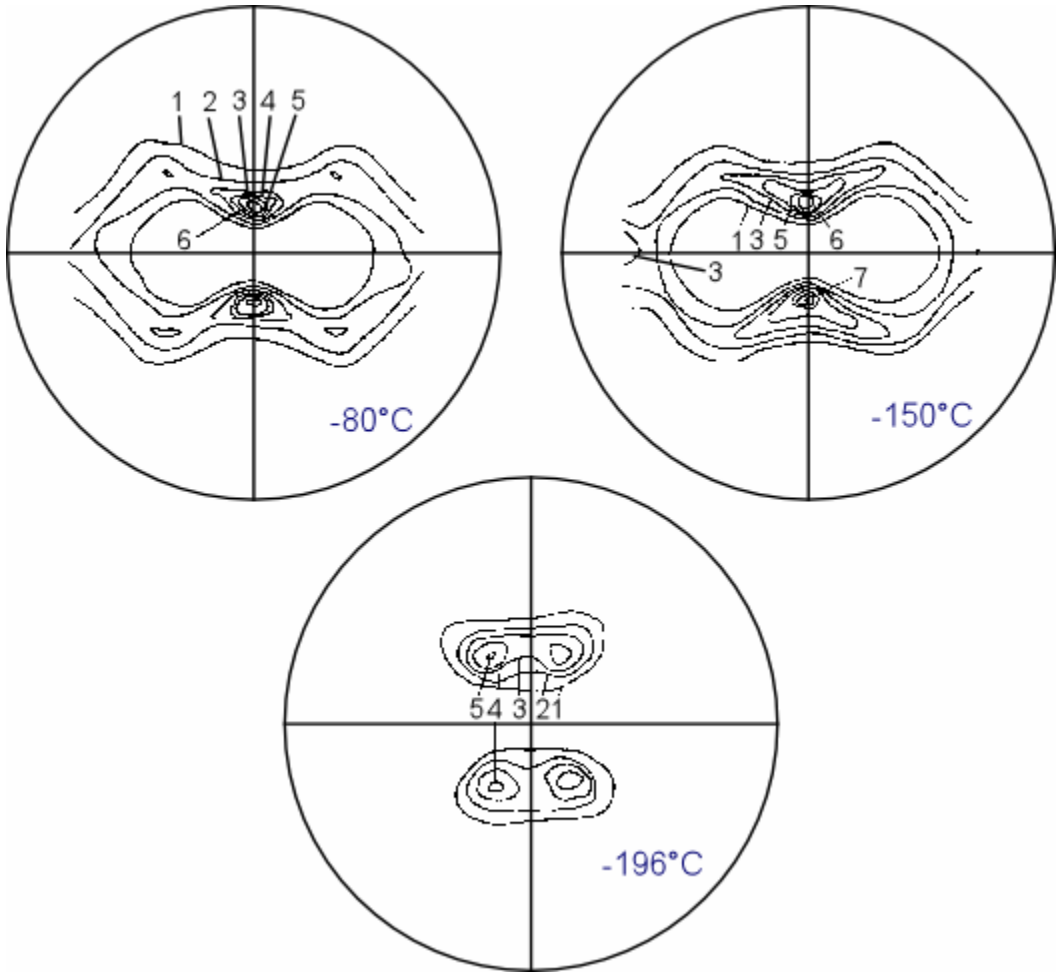
Alloying Effects





Hirsch & Lücke, 1988 , Acta metall. **36**, 2863

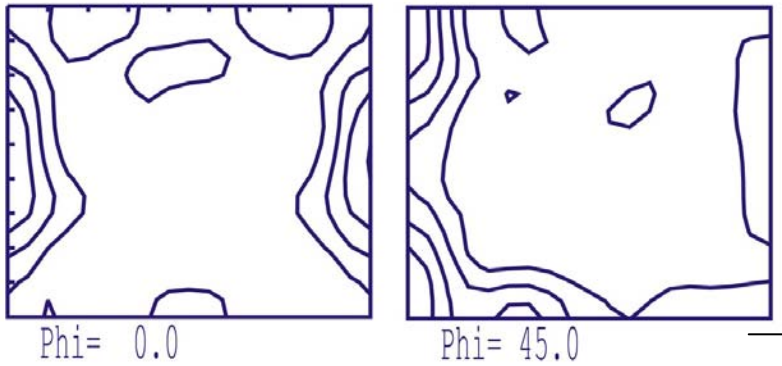
Effect of Temperature

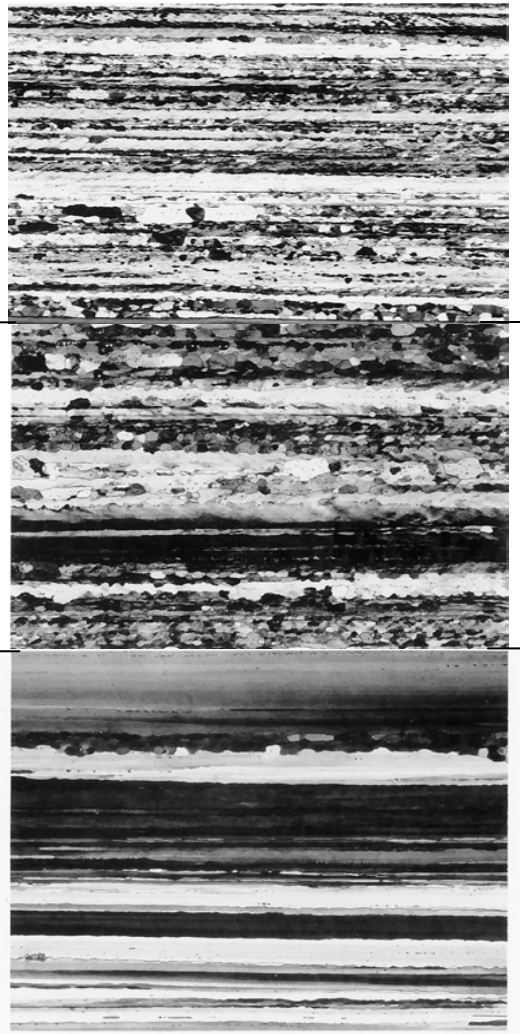


{111} Pole Figures for Pu-9.6 atomic percent Ga rolled to 95% reduction

Moment, R.L. (1972), Rolling texture of Pu-Ga alloys as a function of temperature. *Met. Trans.*, **3**, 1639-1644.

Through Thickness Texture Variation

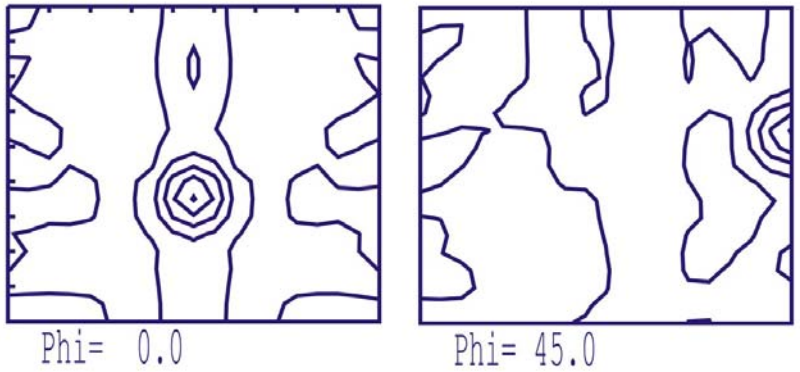




surface

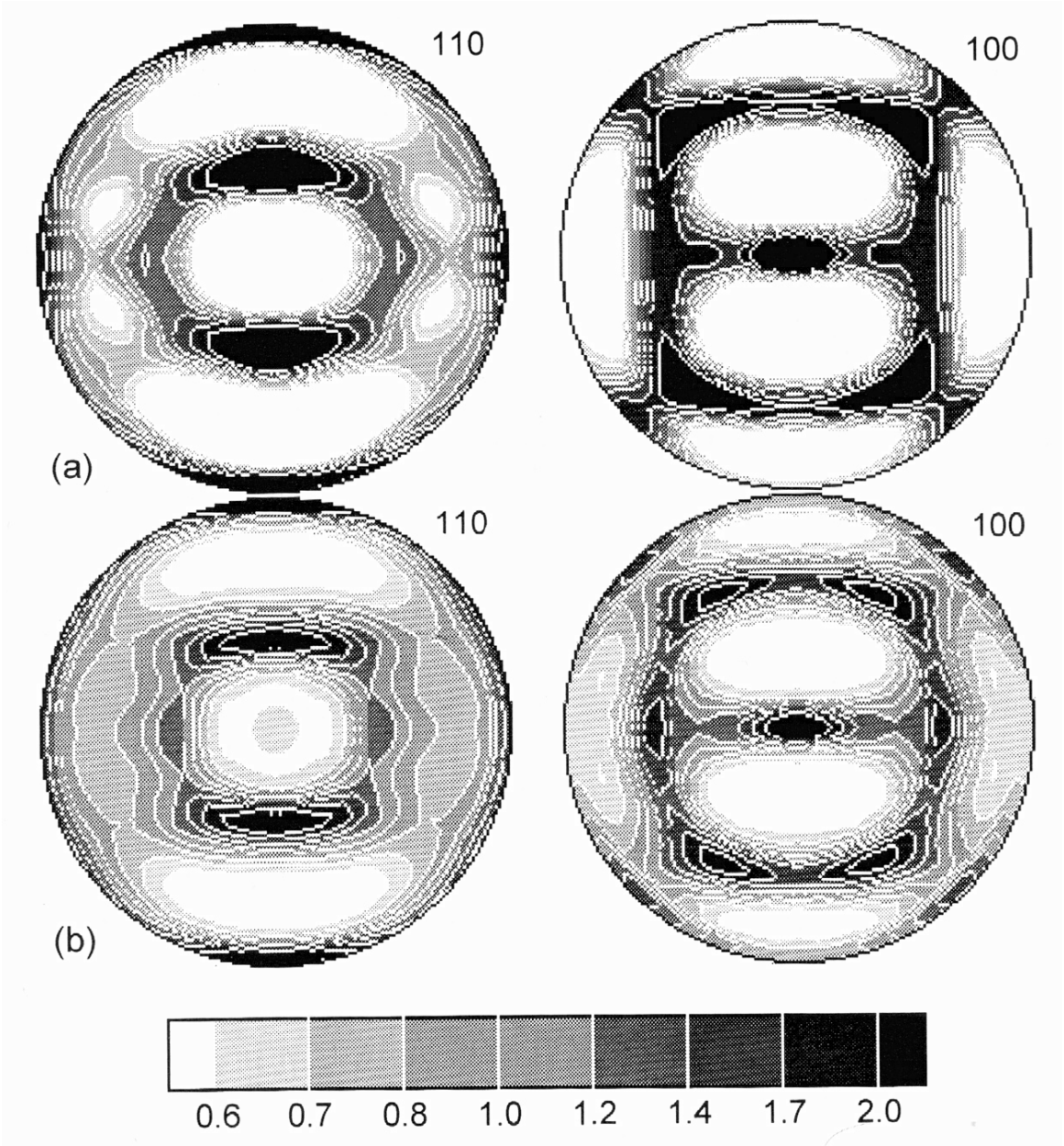
t/4

mid plane



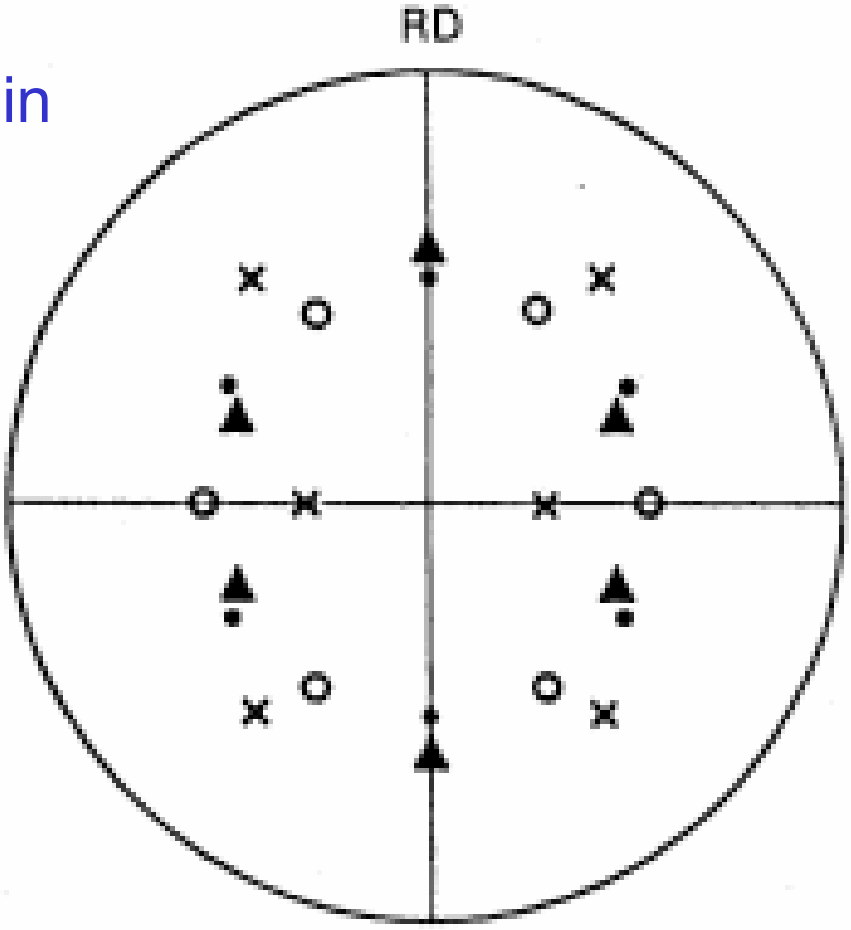
(P. N. Kalu and D. R. Waryoba, LiMAT 2003, Honolulu.)

Typical rolling texture in BCC Materials



{110} and {100} pole figures (equal area projection; rolling direction vertical) for (a) low-carbon steel cold rolled to a reduction in thickness of 80% (approximate equivalent strain of 2); (b) tantalum, unidirectionally rolled at room temperature to a reduction in thickness of 91%.

{100} Pole figure for certain components of rolled bcc metals



Note how very different components tend to overlap in a pole figure.

bcc fibers: the $\phi_2 = 45°$ section

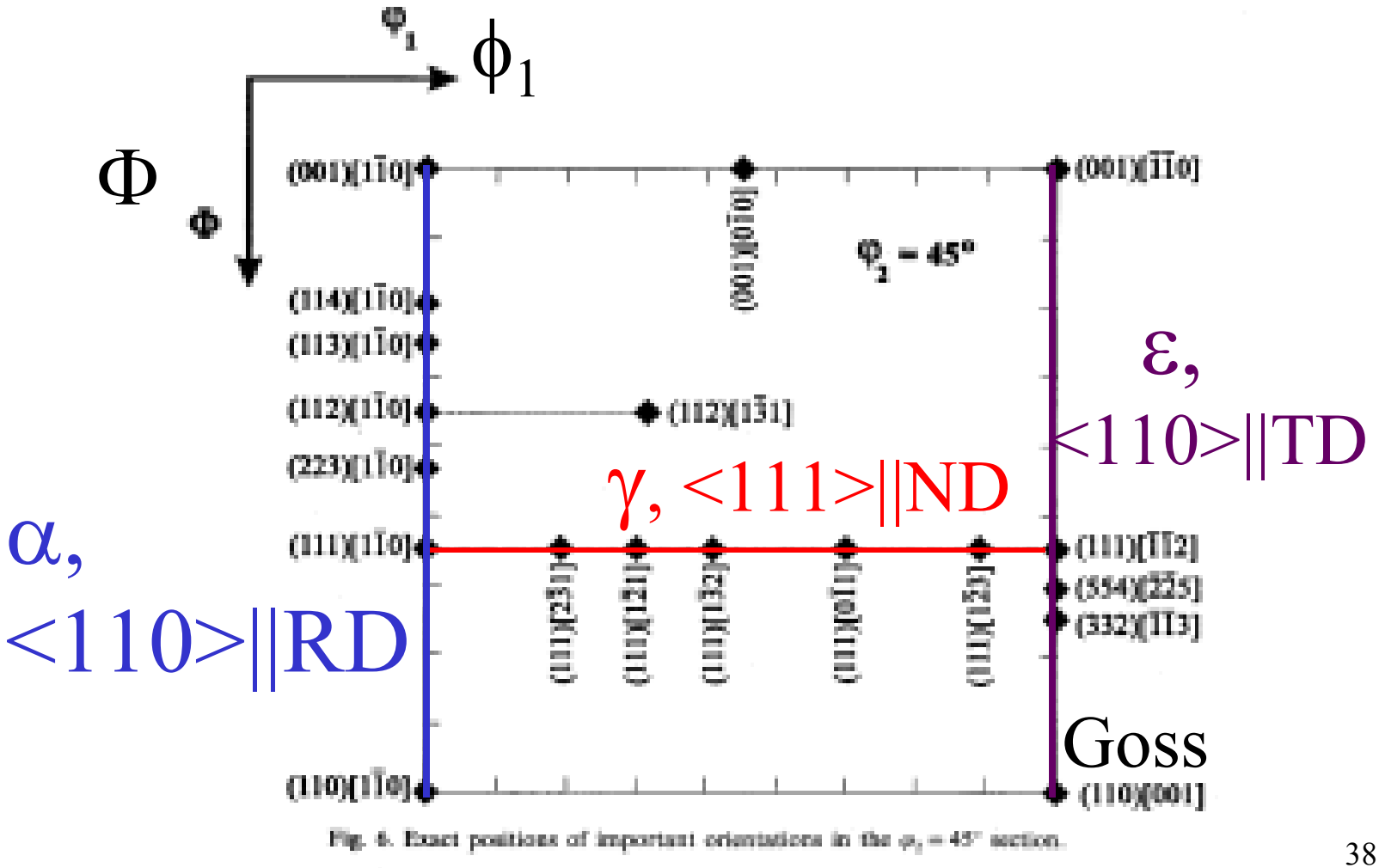


Fig. 6. Exact positions of important orientations in the $\varphi_2 = 45°$ section.

Fe,Fe-Si rolling fiber plots

Note the marked alloy dependence in the alpha fiber; smaller variations in the gamma fiber.

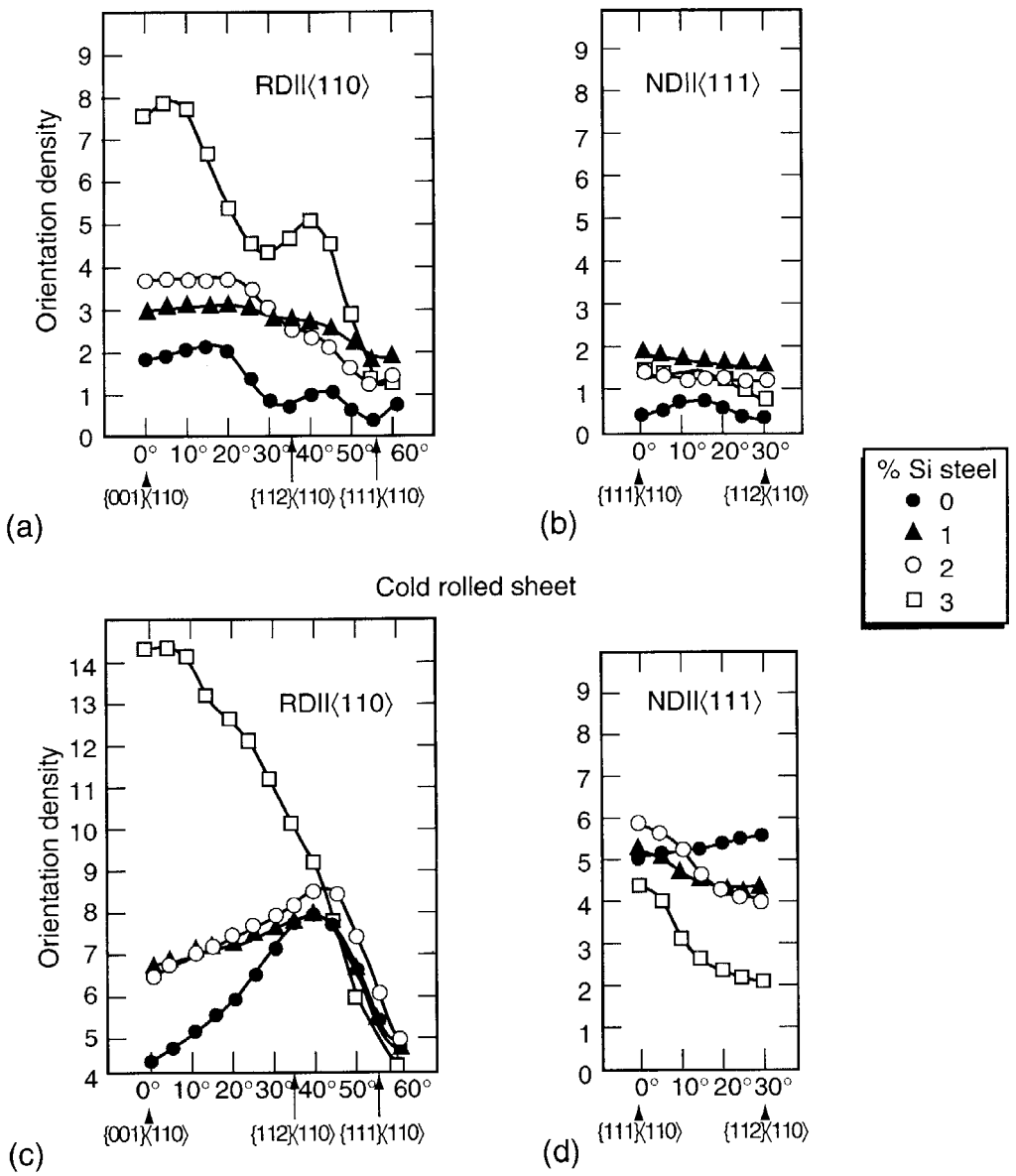
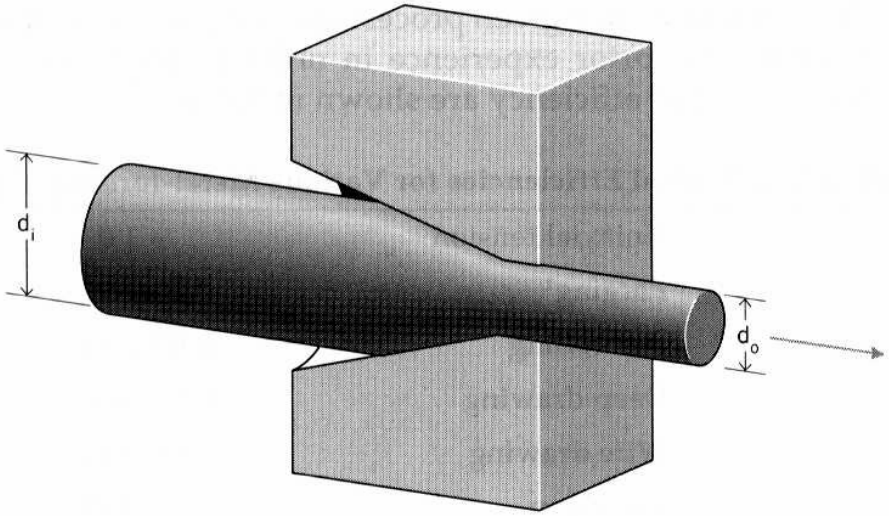
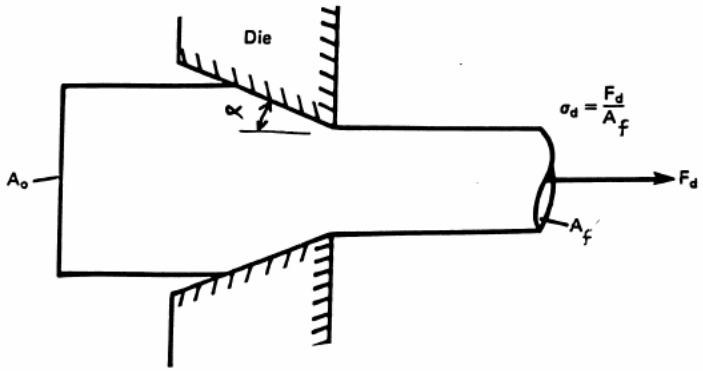


Fig. 17. Plot of the α and γ fibers for a range of iron-Si alloys, including 0, 1, 2, & 3% Si. Increasing silicon leads to stronger α fibers in both the hot-rolled (initial) condition and the cold-rolled condition.

Axisymmetric deformation: Extrusion, Drawing

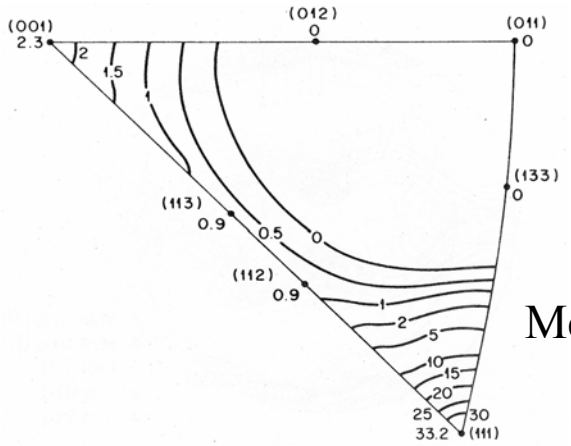




$$\dot{E} = \begin{pmatrix} +\Delta & 0 & 0 \\ 0 & -0.5\Delta & 0 \\ 0 & 0 & -0.5\Delta \end{pmatrix}$$

Axisymmetric deformation

- In fcc metals, axisymmetric deformation (e.g. wire drawing) produces fiber texture: <111> + <100> duplex, parallel to the wire.



McHargue et al., 1959

Schmid and Wassermann (1963): 60% <111> + 40% <100>

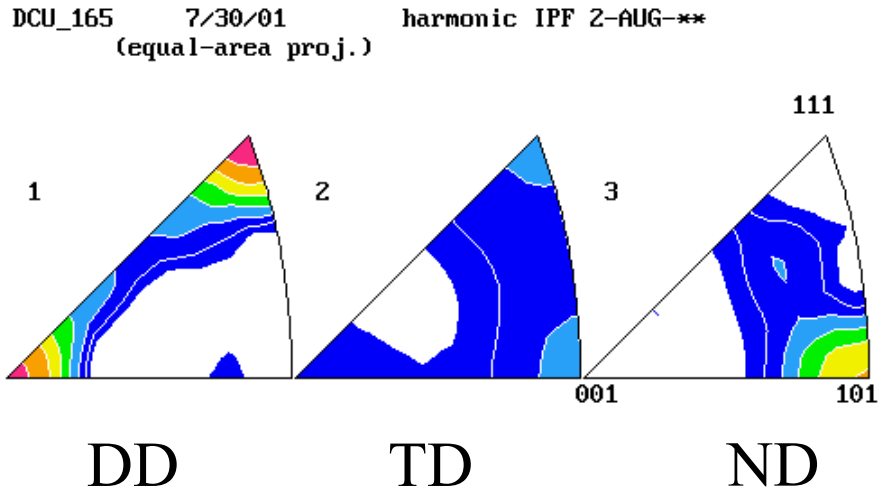
Ahlborn and Wassermann (1963): 66% <111> + 34% <100>

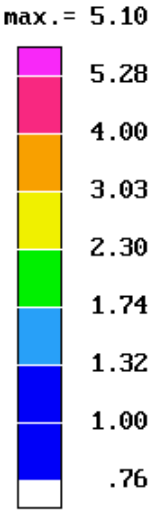
} Electrolytic Copper

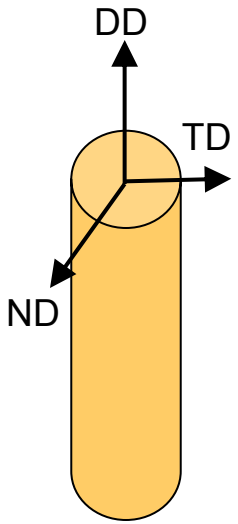
Axisymmetric deformation

- Axisymmetric deformation ~ higher order symmetry, C_∞
- Texture can be represented by an *inverse pole figure* (IPF).
- In IPF, contour lines show the frequency with which the various directions, <uvw>, in the crystal coincide with the specimen axis under consideration

X-ray IPFs showing intensity of crystal directions (u,v,w) parallel to samples axes, DD, TD, and ND of drawn OFHC copper wire

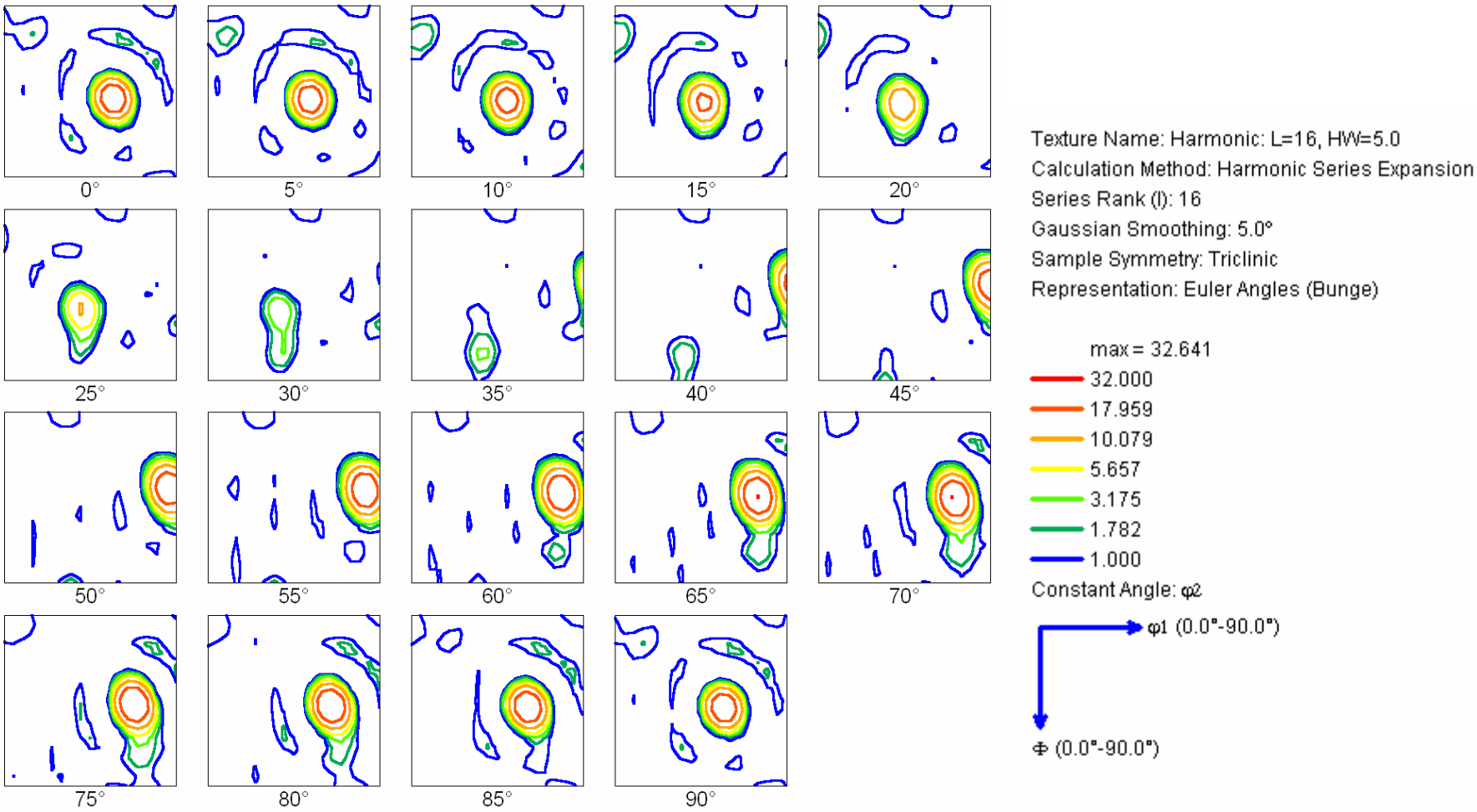






DD – Drawing direction corresponds to RD in rolling

(D. R. Waryoba, Ph. D. Dissertation, FSU, 2003)

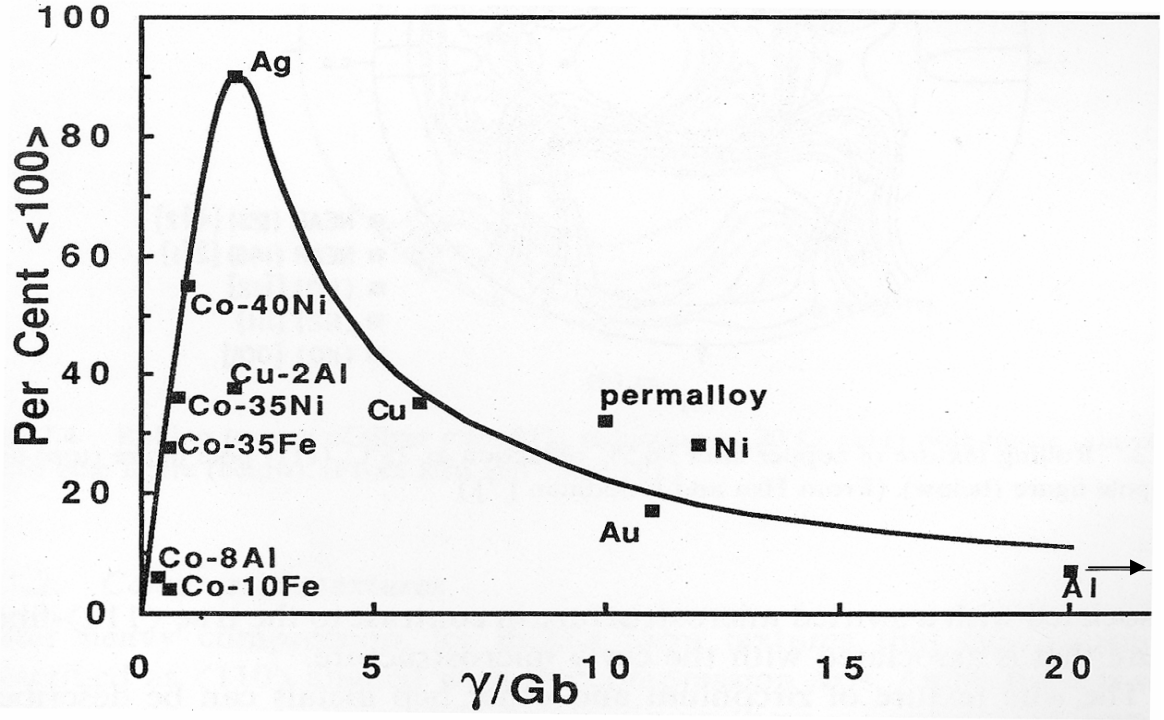


ODF of drawn OFHC copper wire showing high intensity of <111>//DD fiber and weak <100>//DD fiber

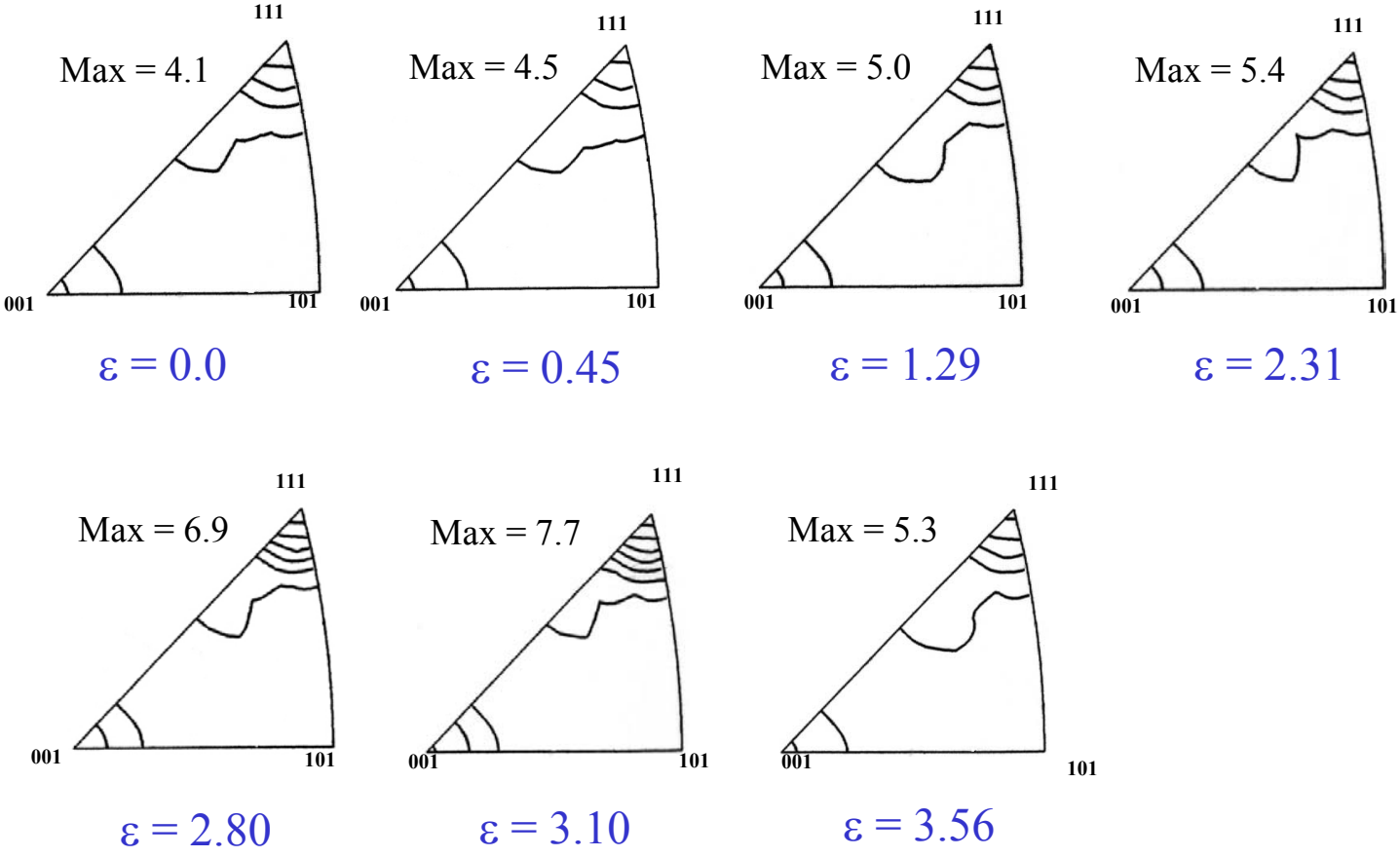
(D. R. Waryoba, Ph. D. Dissertation, FSU, 2003)

Axisymmetric deformation

- The relative proportions of the two components are determined by the stacking fault energy [English et al., 1965] and vary in a complex manner.



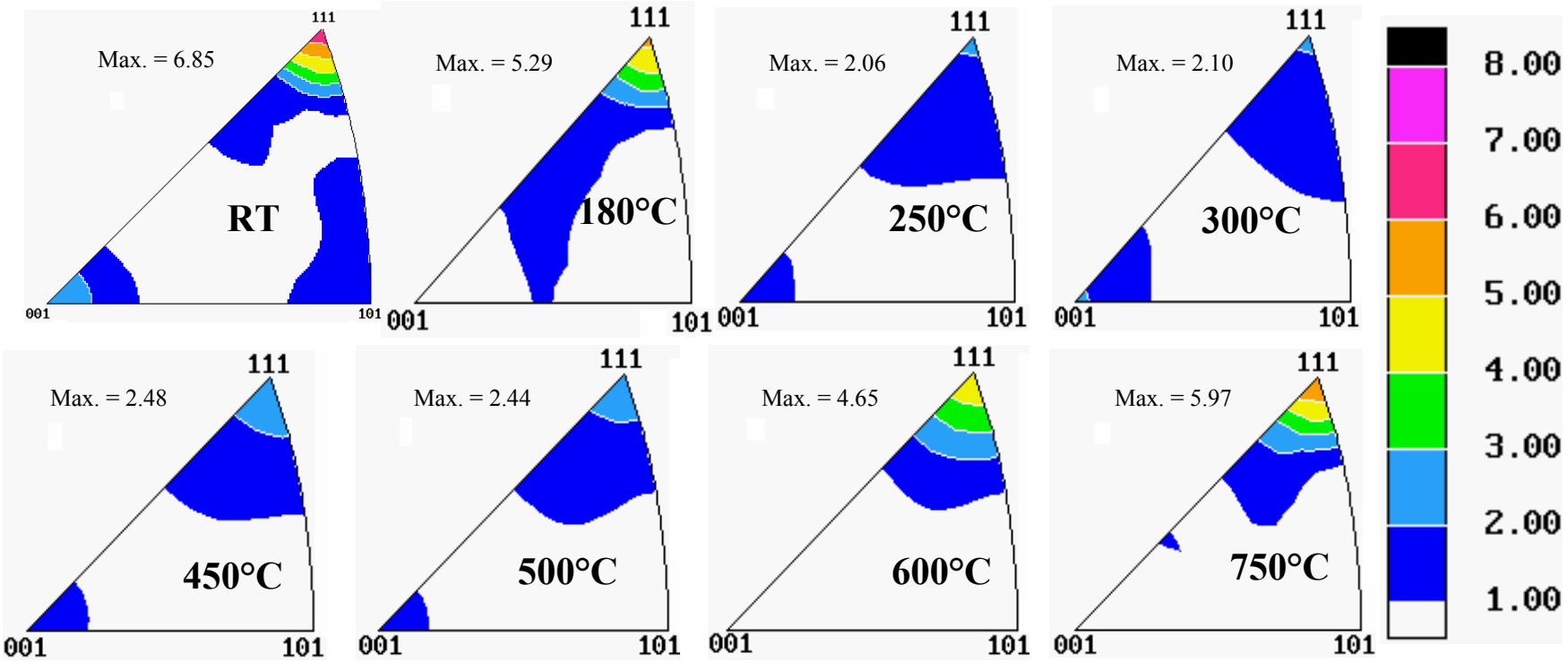
Effect of deformation strain



X-ray IPFs showing the effect of strain on the texture of OFHC copper wire

(D. R. Waryoba, Ph. D. Dissertation, FSU, 2003)

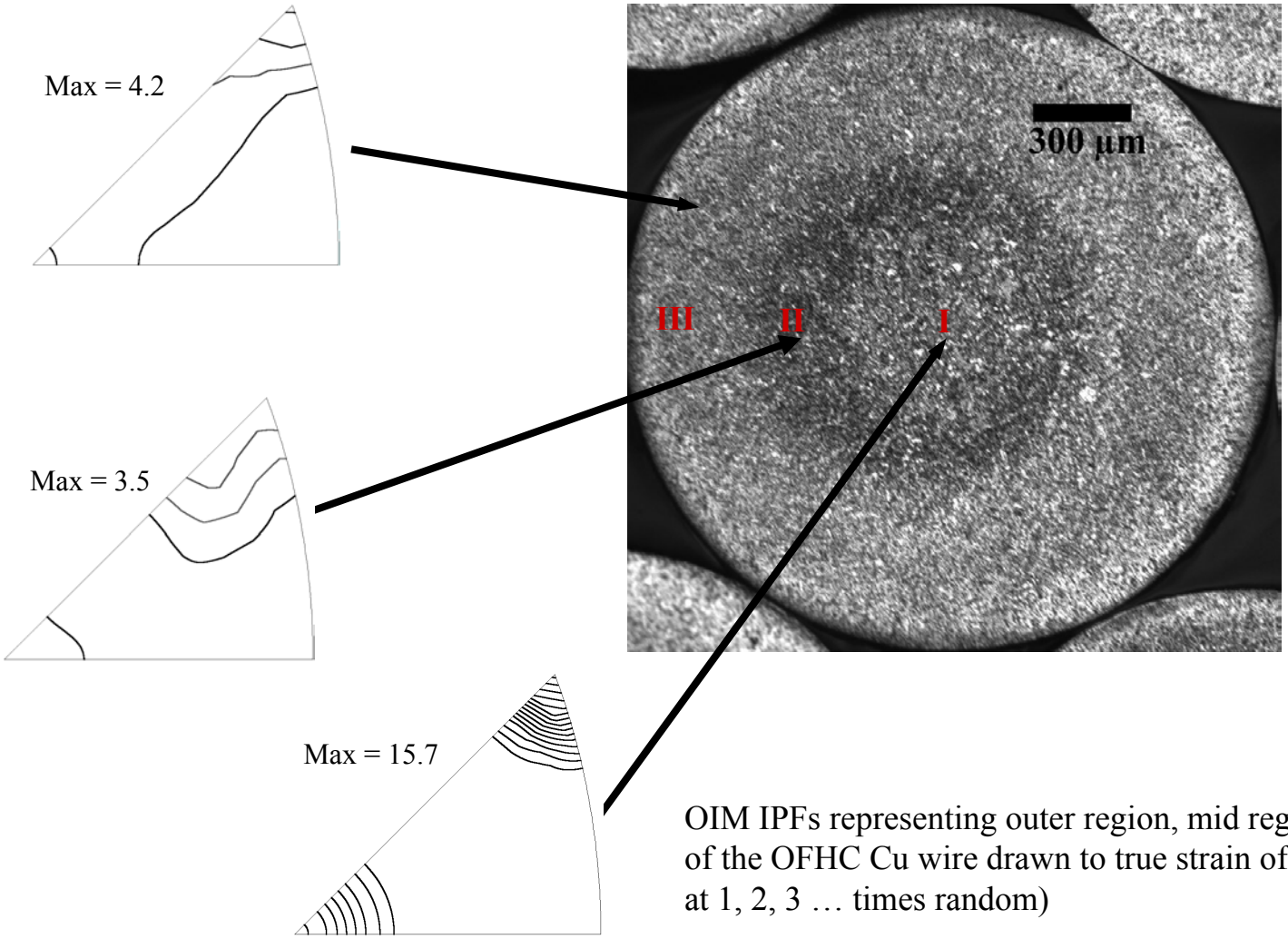
Effect of Temperature



X-ray IPFs showing the effect of annealing temperature on the texture of OFHC copper wire, initially drawn to true strain of 2.31

(D. R. Waryoba and P. N. Kalu, TMS 2003,San Diego, CA.)

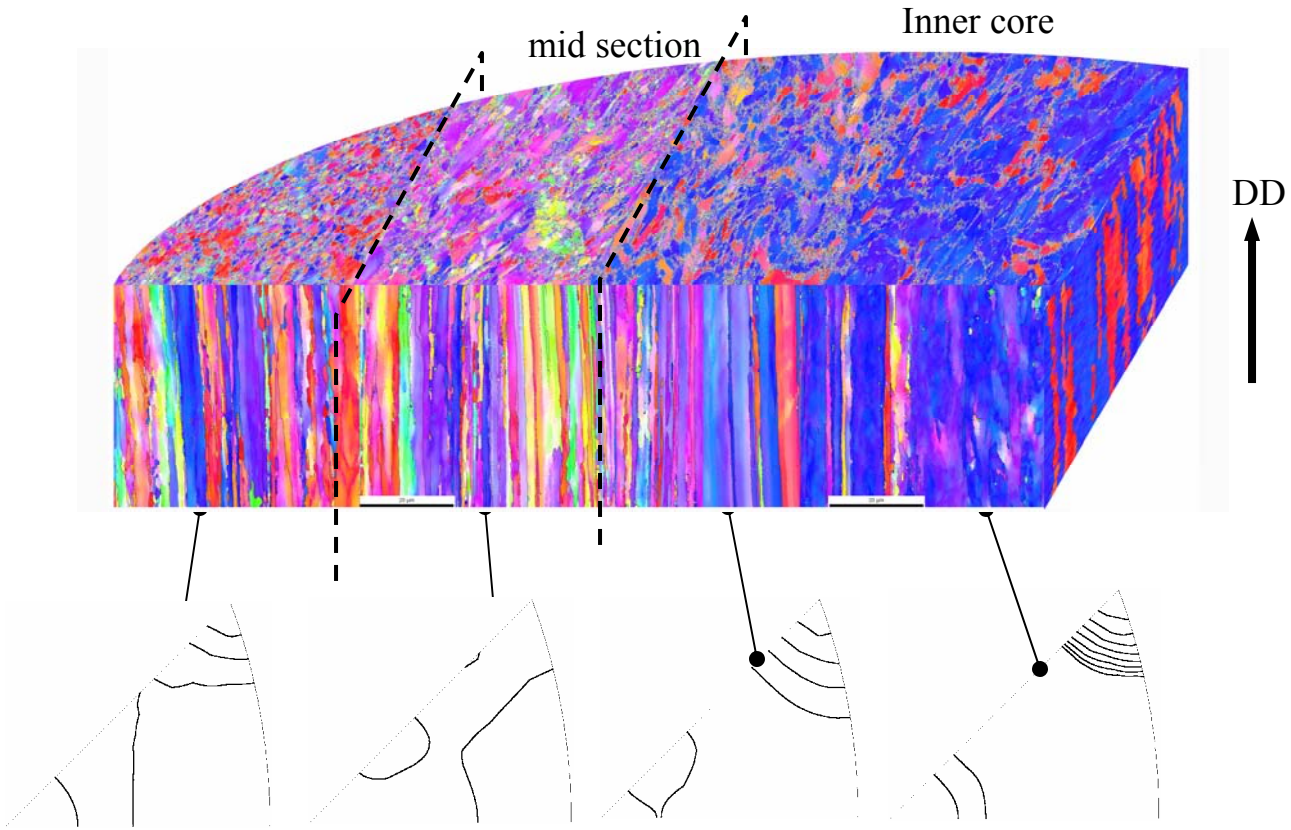
Texture inhomogeneity in Drawn Wires



OIM IPFs representing outer region, mid region, and inner core of the OFHC Cu wire drawn to true strain of 2.31 (contours are at 1, 2, 3 … times random)

*(Waryoba, D. R., and Kalu, P. N., Mater. Sci. Forum **495-497** (2005), 895-900)*

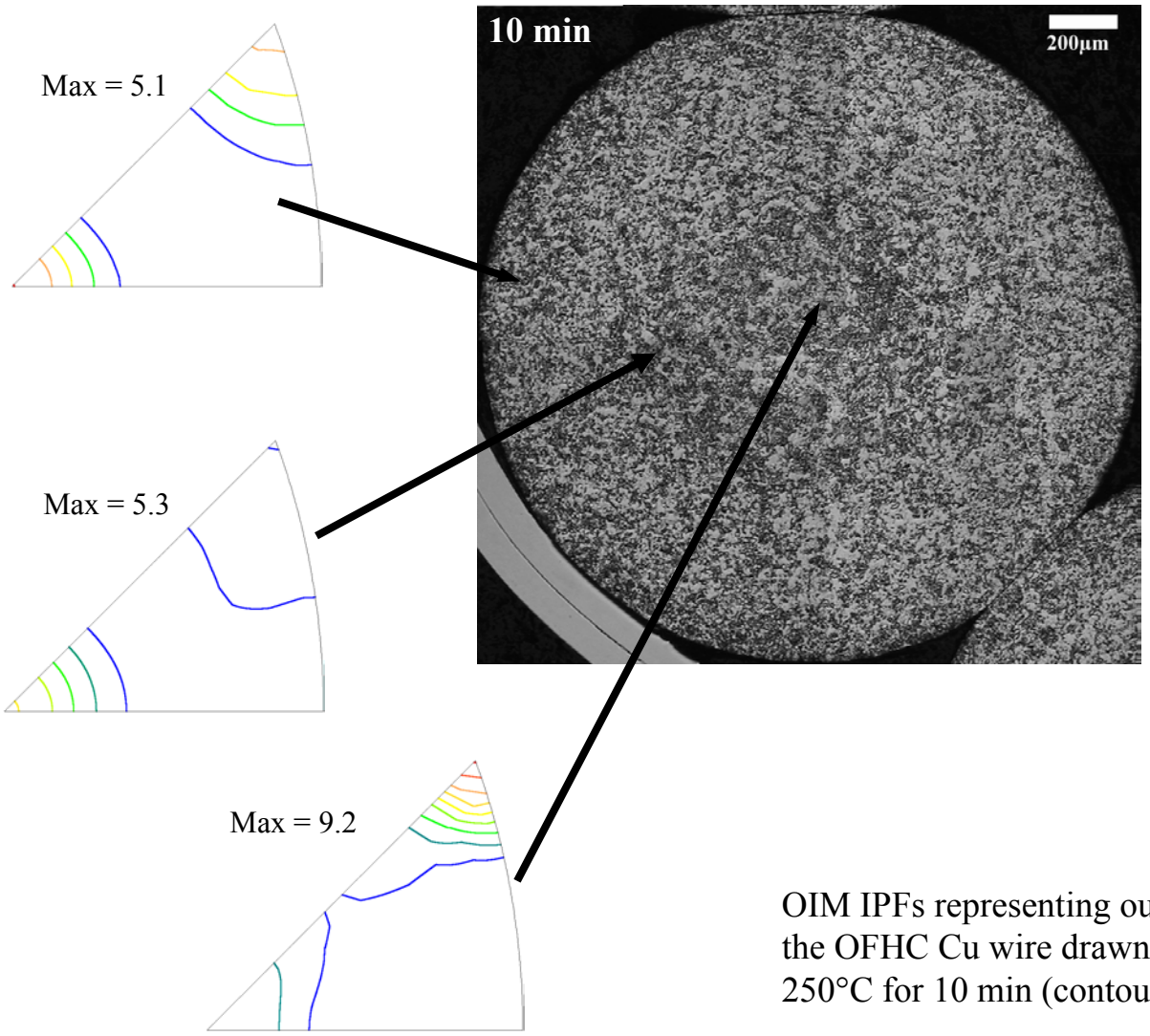
Texture inhomogeneity in Drawn Wires



[contours at 1, 2, 3, … times random]

(Waryoba, D. R., and Kalu, P. N., Mater. Sci. Eng. (2006), In Press)

Texture inhomogeneity in Drawn Wires



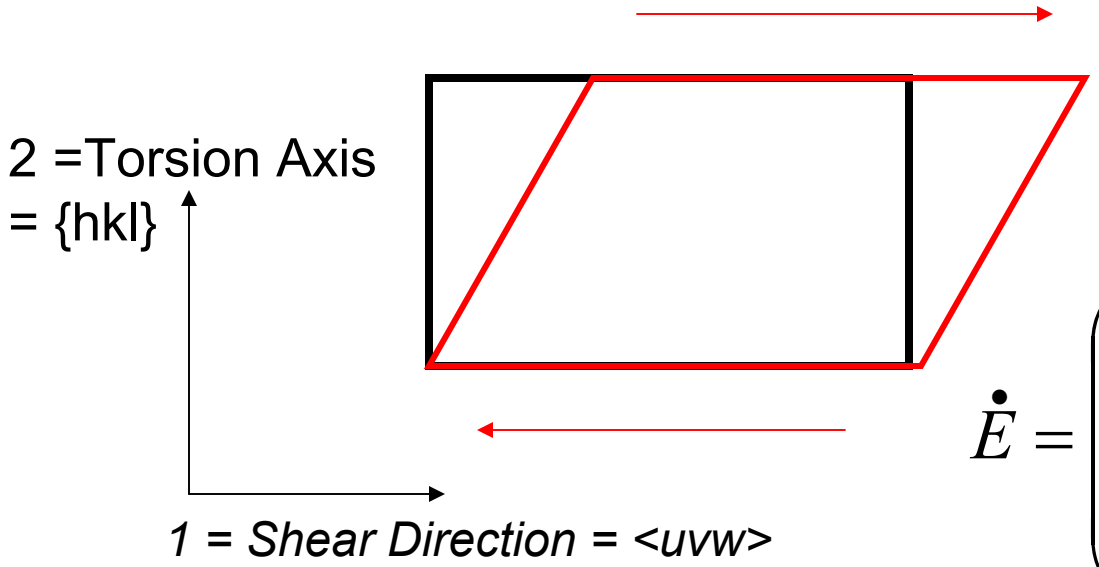
OIM IPFs representing outer region, mid region, and inner core of the OFHC Cu wire drawn to true strain of 2.31 and annealed at 250°C for 10 min (contours are at 1, 2, 3 … times random)

*(Waryoba, D. R., and Kalu, P. N., Mater. Sci. Forum **495-497** (2005), 895-900)*

Shear Texture

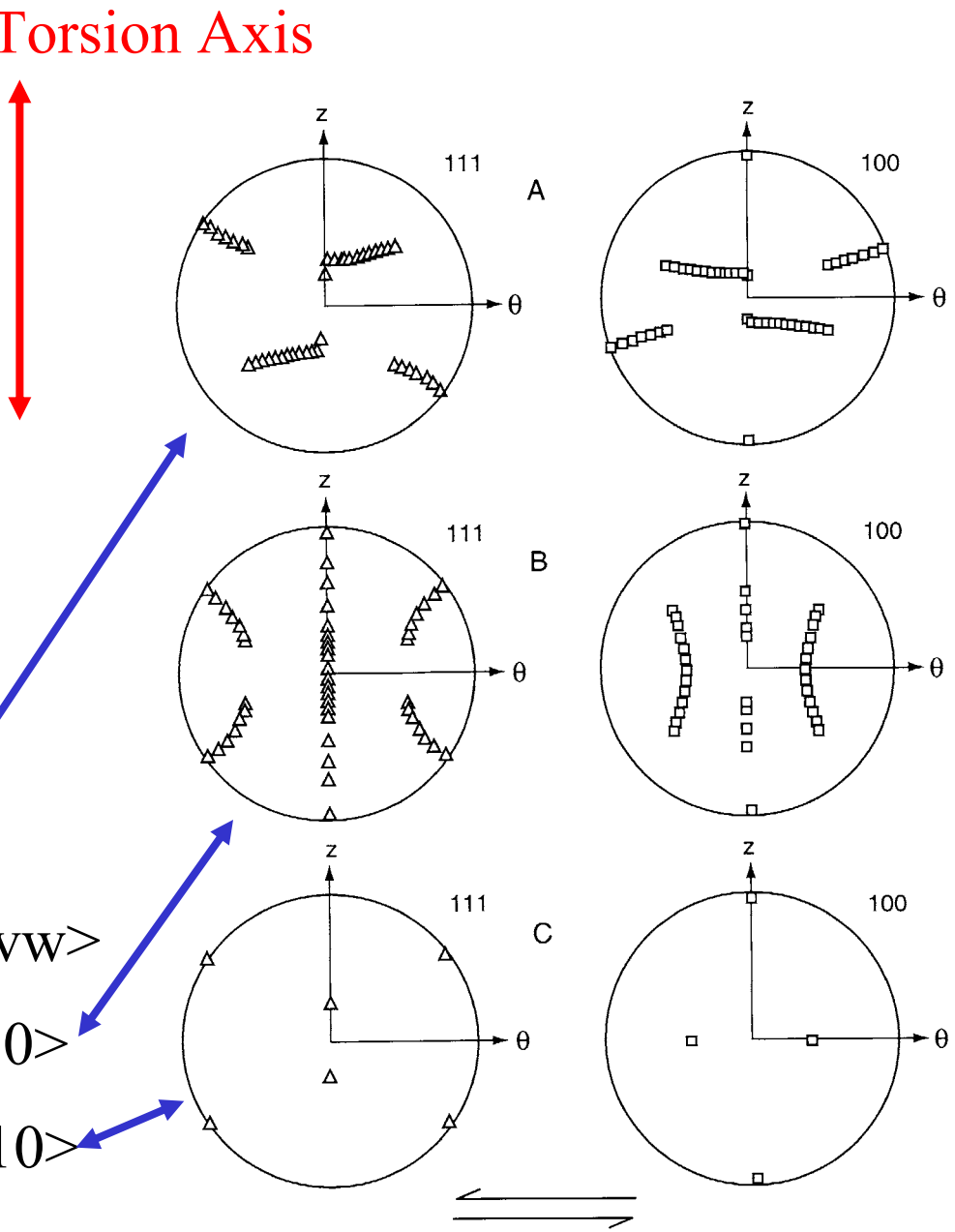
- Shear strain means that displacements are tangential to the direction in which they increase.
- Shear direction=1, Shear Plane ⊥ 2-axis

$$F = \begin{pmatrix} 1 & \gamma & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



$$\dot{E} = \begin{pmatrix} 0 & +\Delta & 0 \\ +\Delta & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Idealized shear texture components for fcc metals



A partial fiber {111}<uvw>

B partial fiber {hkl}<110>

C component: {001}<110>

Shear Texture Components

- Why study shear textures? Shear strain is common near the surface of rolled parts, for example.
- Partial Fibers: Components

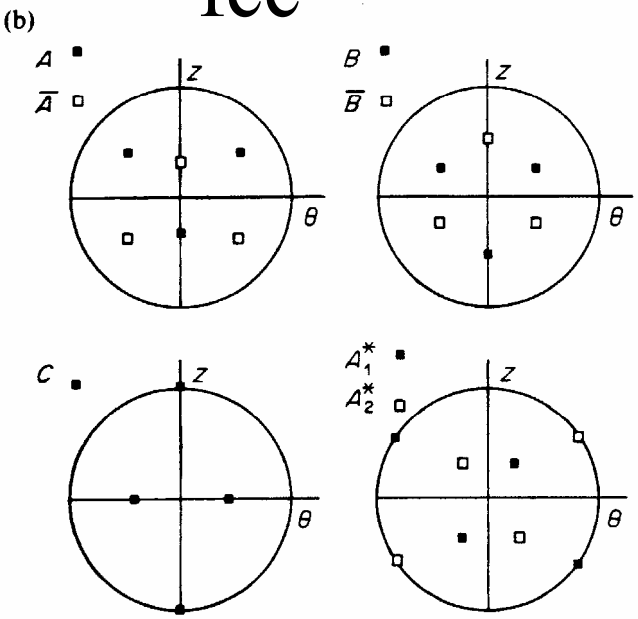
A/D	{111}<uvw>…<110>
B	{hkl}<110> … {112}
C	{001}<110>
D	{112}<111>
E	{011}<111>
F	{110}<001>

Montheillet et al., Acta metall., **33** 705, 1985

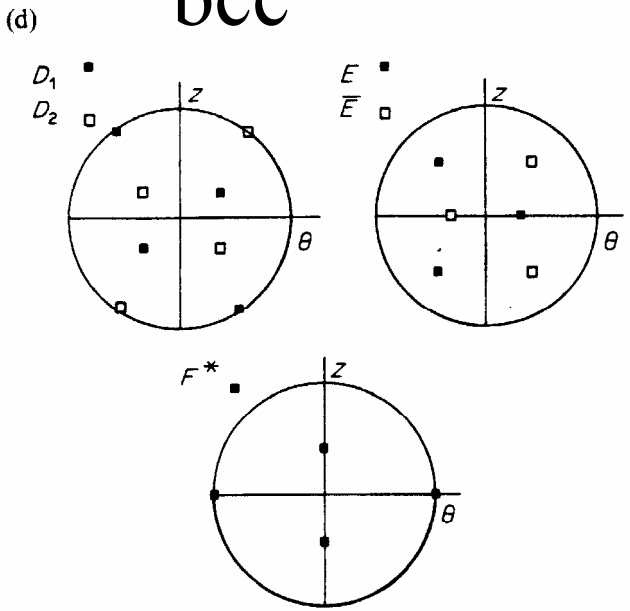
Table 1. Notation and Miller indices used for the different ideal orientations

A	$\{1\bar{1}1\}\langle 110\rangle$	C	$\{001\}\langle 110\rangle$
$\bar{A}$	$\{\bar{1}1\bar{1}\}\langle\bar{1}\bar{1}0\rangle$	D_1	$\{1\bar{2}1\}\langle 111\rangle$
A_1^*	$\{\bar{1}\bar{1}1\}\langle 112\rangle$	D_2	$\{\bar{1}\bar{1}2\}\langle 111\rangle$
A_2^*	$\{11\bar{1}\}\langle 112\rangle$	E	$\{0\bar{1}1\}\langle 111\rangle$
B	$\{1\bar{1}2\}\langle 110\rangle$	$\bar{E}$	$\{01\bar{1}\}\langle\bar{1}\bar{1}\bar{1}\rangle$
$\bar{B}$	$\{\bar{1}1\bar{2}\}\langle\bar{1}\bar{1}0\rangle$	F^*	$\{110\}\langle 001\rangle$

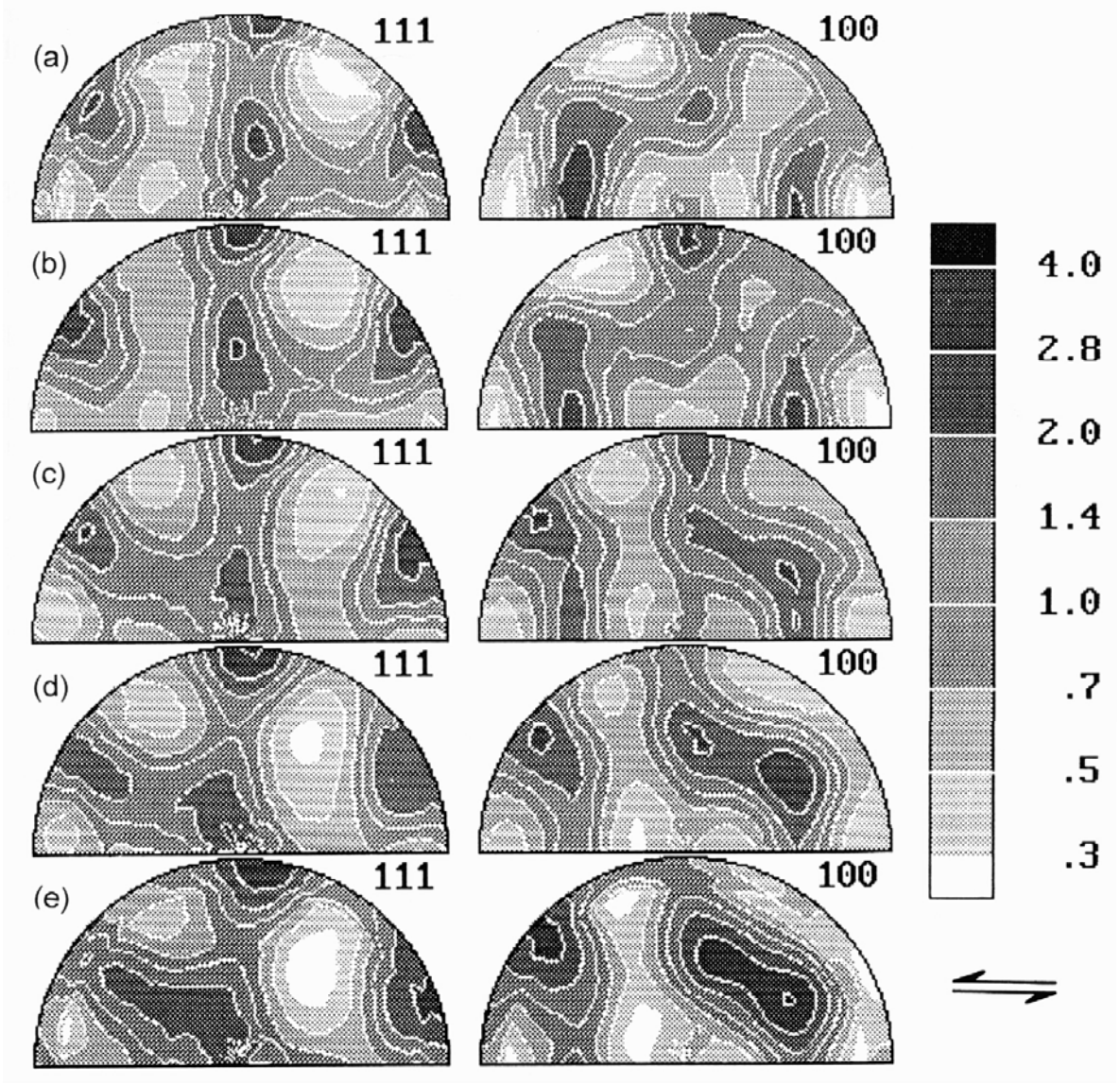
fcc



bcc



FCC Torsion Textures

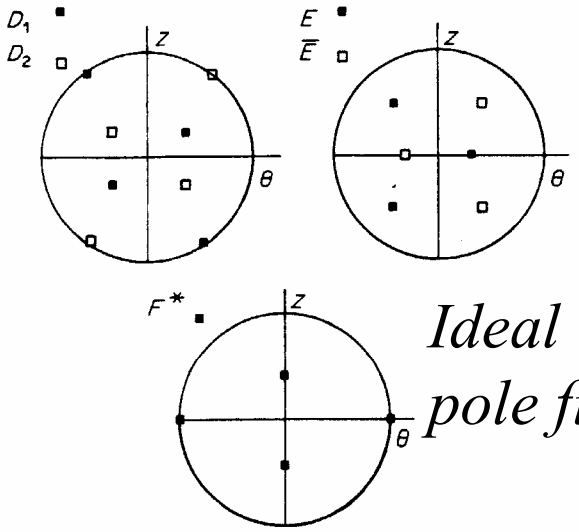


Plots of {111} and {200} pole figures (equal area projection; torsion axis vertical) for the following materials deformed in torsion; the shear direction points to the left in these figures.

a) Nickel at γ=3.6
b) Copper at γ=3.5
c) Silver at γ=3.5
d) Cu-30Zn at γ=3.5
e) Ni-60Co at γ=3.2

Note that the partial "A" fiber is present in Ni and Cu, but is absent in the other materials. Silver, brass and Ni-60Co show instead a "D" fiber which is similar to the A fiber but rotated approximately 90° about the torsion axis. The B fiber is present to varying degrees in all the materials.

bcc torsion textures: Fe



Ideal {100} pole figures

Texture dominated by {112}<111> and {110}<001>

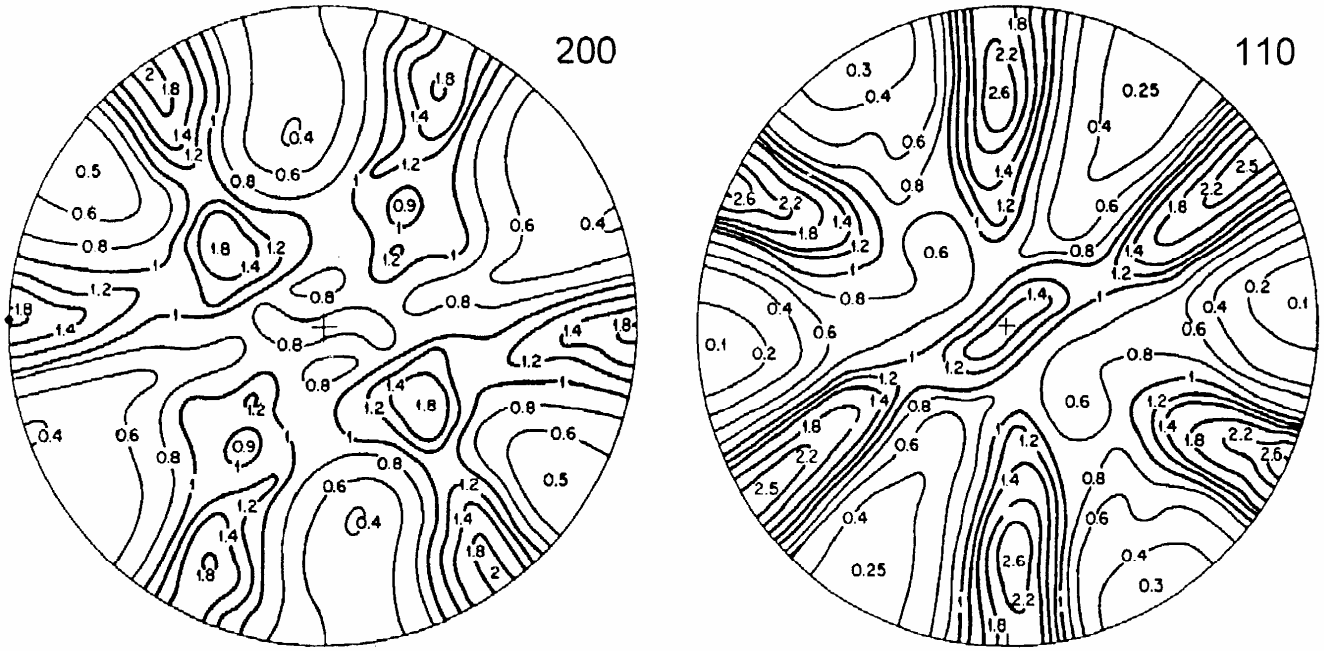
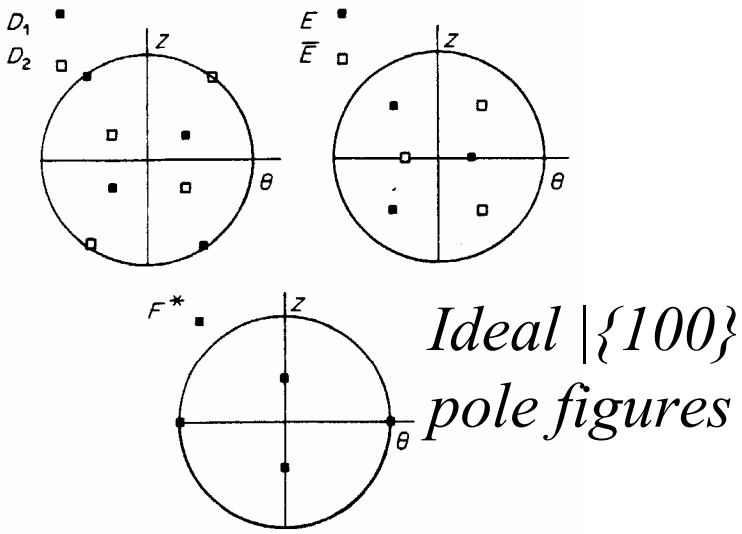


Fig. 18. Experimental 200 and 110 pole figures for Armco iron sheared to γ=2.1 (ε_{vM}=1.2) [WILLIAMS 1962] (Stereographic projection.) The shear direction points right on top.

bcc torsion textures: Ta

(a) initial texture from swaged rod;
(b) torsion texture



Ideal {100} pole figures

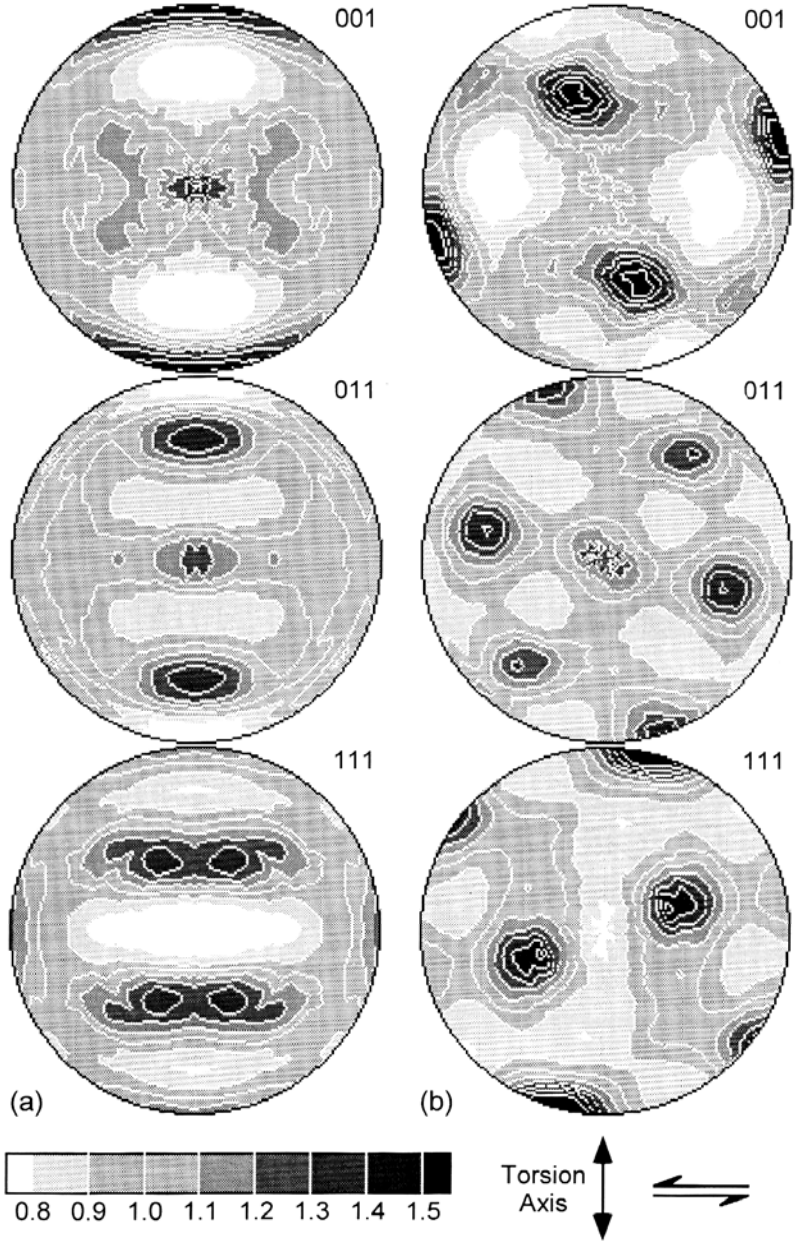
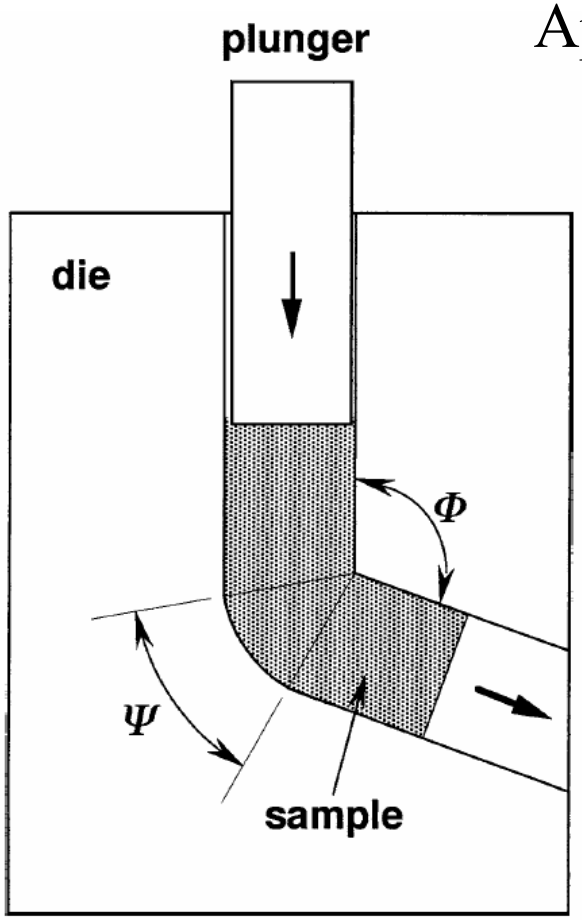


Fig. 19. Recalculated 111, 100 and 110 pole figures for tantalum: (a) initial texture; (b) tested in torsion to ε_{vM}=1.4. Equal-area projection.

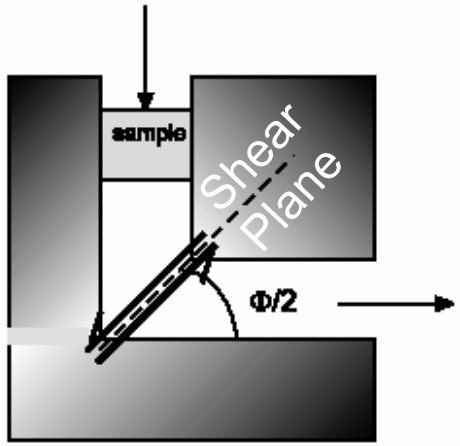
Equal Channel Angular Extrusion



Approximation: Simple shear deformation

$$\gamma_N = N\left[2\cot\left(\frac{\Phi}{2}+\frac{\Psi}{2}\right)+\Phi\,\mathrm{cosec}\left(\frac{\Phi}{2}+\frac{\Psi}{2}\right)\right]$$

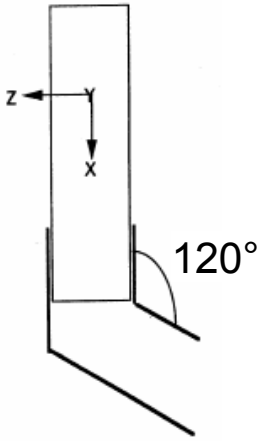
$$\varepsilon_N = \frac{\gamma_N}{\sqrt{3}}$$

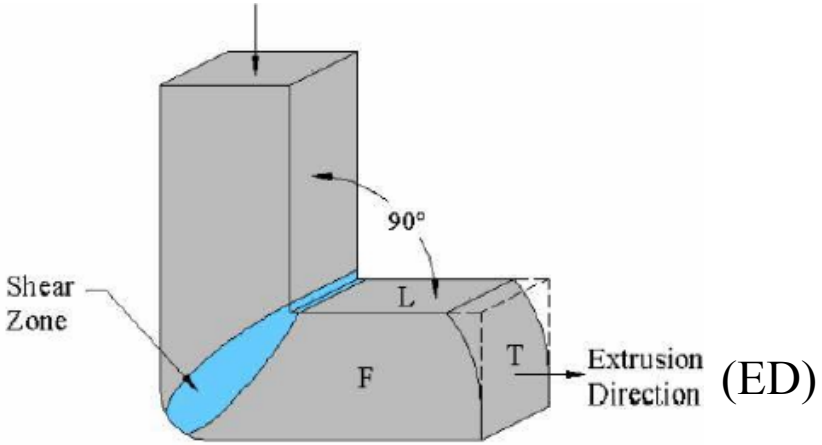


$$\varepsilon_N \approx \frac{2\cot(\Phi/2)}{\sqrt{3}} = 1.15N$$

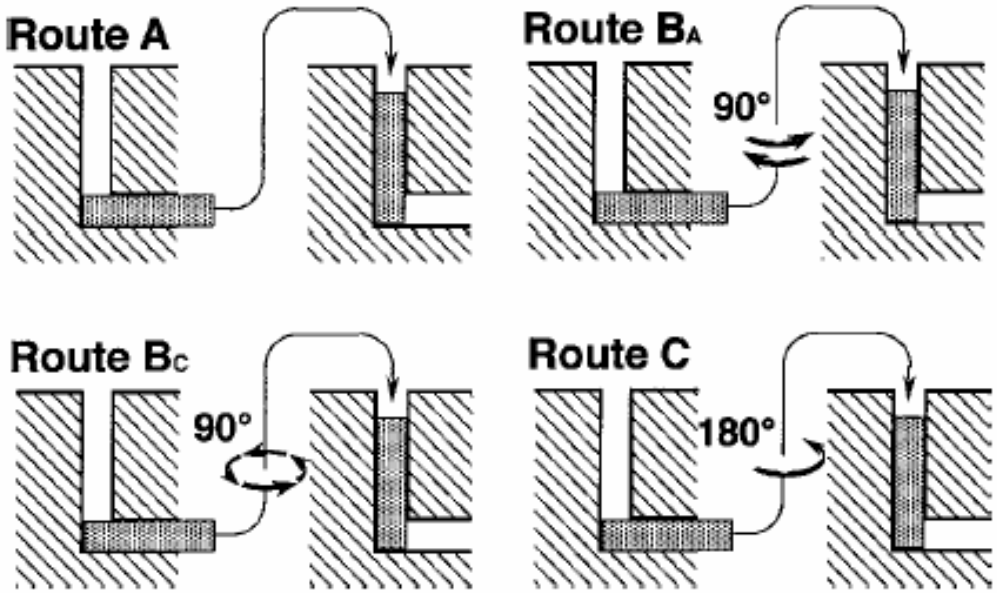
V. M. Segal, *Mater. Sci. Eng.* A197:157-164 (1995).

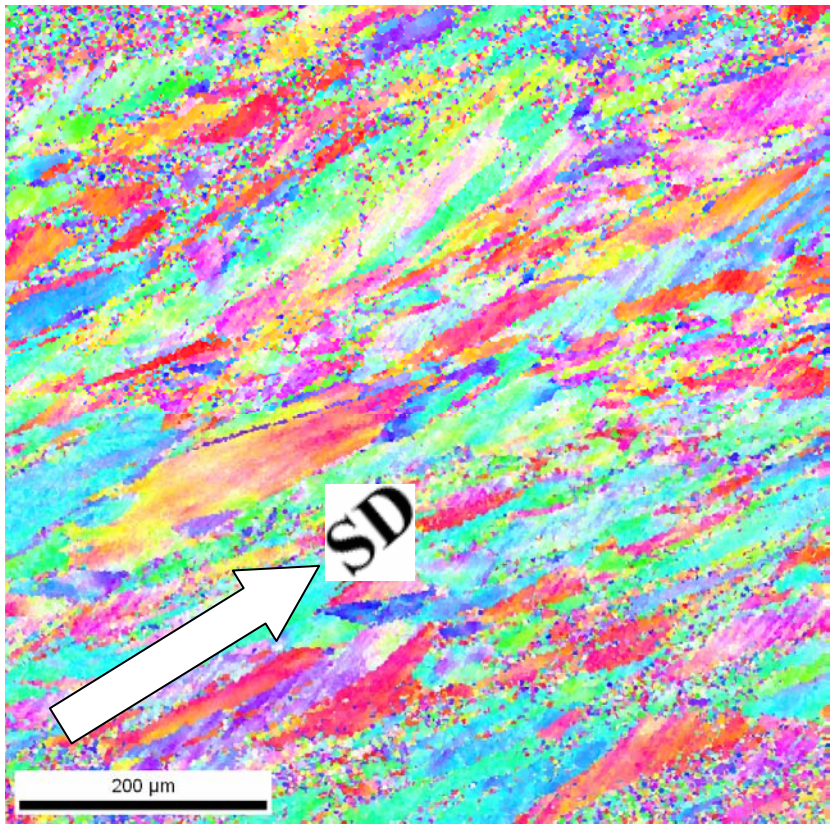
Die Angle





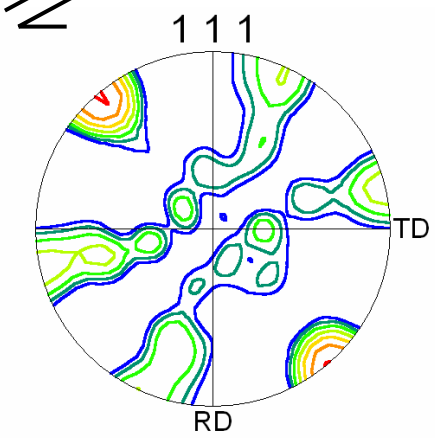
Processing routes

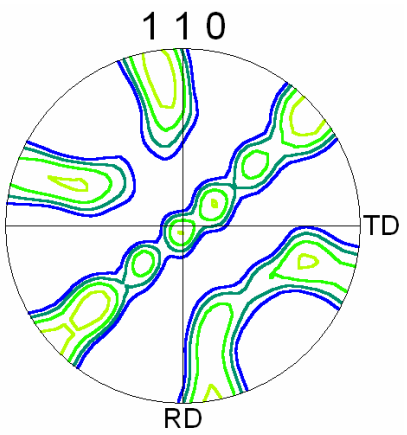


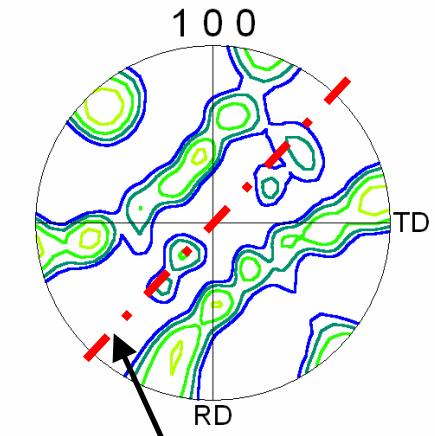


ED

Simple shear



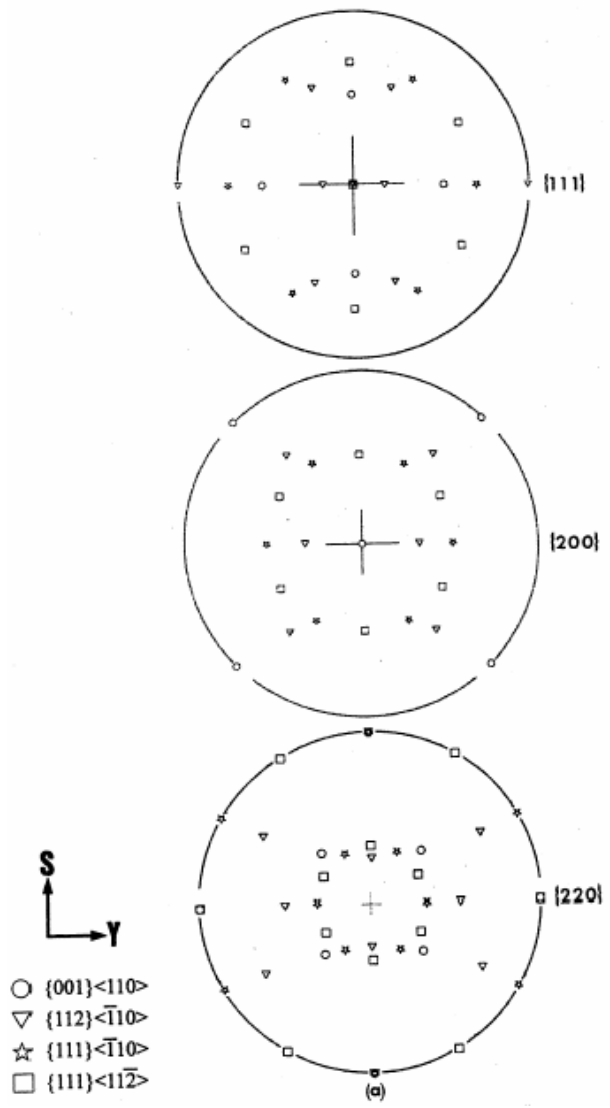




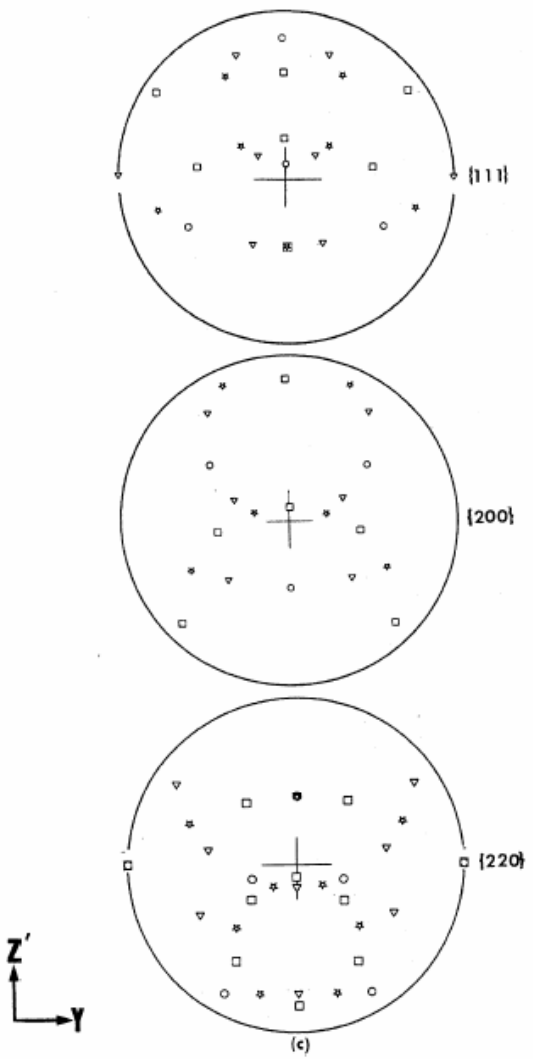
ED

Shear direction (SD)

OFHC Copper, processed to 1 pass through a 90° channel.
(D. R. Waryoba and P. N. Kalu)



Torsional Ideal orientations



After 45° rotation about the Y-axis

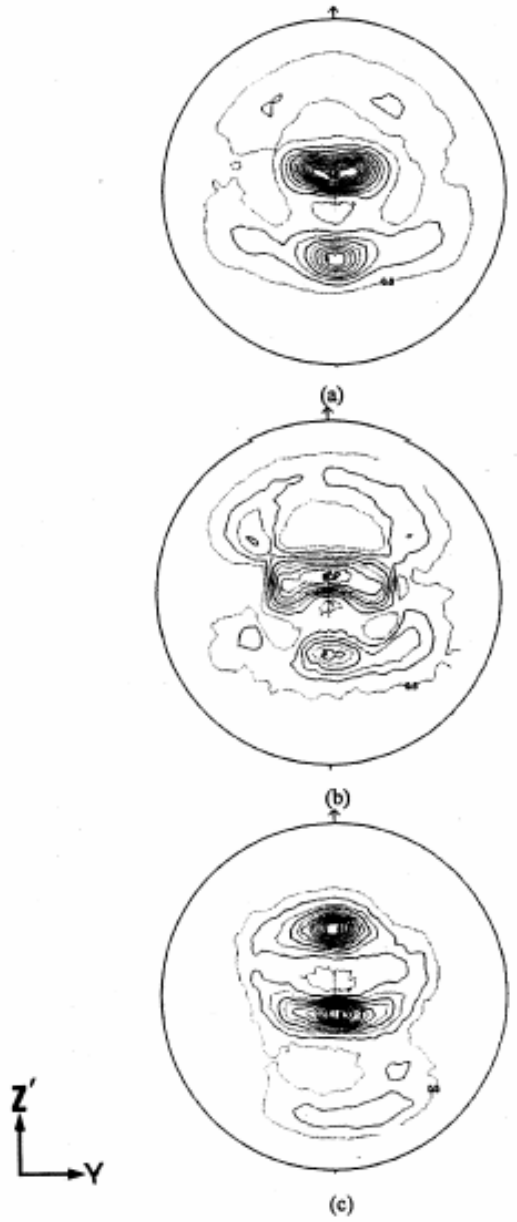


Fig. 4. (a) {111}, (b) {200}, and (c) {220} pole figures of copper deformed by ECAE with the 90° die.

Summary

- Typical rolling, axisymmetric and torsional textures illustrated for *fcc* and *bcc* metals as a function of alloy type (stacking fault energy), deformation character (strain type) and temperature.
- For restricted glide, *bcc* textures are transposition of *fcc* textures
- Pole figures are recognizable for standard deformation histories but orientation distributions provide much more detailed information.